Internalizing Environmental Risk: Insurance Design and Firm Behavior in Hazardous Industries

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Public trust funds in thirty-eight U.S. states insure underground storage tank (UST) leaks at a uniform flat fee, even though leak risk varies widely with observable traits such as tank age and wall construction. This paper quantifies how replacing flat-fee pooling with actuarially priced coverage changes firm behavior and environmental outcomes. I exploit Texas's 1999 closure of its trust fund, which forced all owners into a private market that sets premiums according to facility risk. A facility-year panel covering 1990–2021 links EPA inventories with administrative microdata for Texas and eighteen control states that retained public funds. I construct a make-model comparison sample that matches Texas facilities to control-state counterparts on the observable risk characteristics, tank age bin and wall construction type, that private insurers price, ensuring the comparison group is balanced on the dimensions along which the treatment effect is theoretically heterogeneous. Difference-in-differences and event-study estimates show that risk-rated premiums increased annual closure rates by two to three percentage points, shifted closures from market exit to on-site replacement with safer equipment, and reduced the probability of a reported leak by about one percentage point, nearly the entire pre-policy baseline for the high-risk single-wall cohort. The treatment effect is concentrated among older and single-walled facilities, consistent with the premium-gradient mechanism, and is near zero for young double-walled tanks. The findings provide the first causal micro-evidence that cost-based environmental liability insurance can internalize pollution risk more effectively than uniform public pooling, offering clear guidance for states that are reconsidering the structure of UST insurance programs.

Introduction

Underground storage tanks holding motor fuels and other hazardous liquids sit beneath about half a million U.S. facilities, and leaks from these tanks are the leading documented source of groundwater contamination EPA (2024). Cleanup is costly; median remediation expenses exceed $100 000 per site Marcus (2021), and the probability of a release rises sharply with observable characteristics such as tank age and whether the walls are single or double. Despite these predictable differences, thirty-eight states finance remediation through public trust funds that charge every facility the same premium, diluting incentives to invest in safer technology and crowding out private insurers (**?**).

Economic theory predicts that tying premiums to expected damages should induce firms to internalize environmental risk (Shavell 1984c; Polinsky 1980), yet the relevant margin here is not liability per se but the rating regime under which liability is financed. Uniform-premium public funds operate as community-rated pools: every covered facility pays the same fee regardless of its observable risk characteristics, replicating the cross-subsidization and incentive dilution that Pauly (1974) identified in flat-rate insurance and that Rothschild and Stiglitz (1976) showed prevents competitive separating equilibria. Under community rating, two distortions operate simultaneously. Adverse selection pushes high-risk facilities toward the subsidized pool and crowds private risk-rated alternatives out of the market. Moral hazard suppresses prevention investment because the marginal high-risk facility bears no incremental premium from operating leak-prone equipment. Empirical tests of this mechanism for USTs are scarce. Michigan's mid-1990s shift toward private, risk-rated coverage suggested large gains but relied on aggregate counts that could not isolate facility-level responses (Yin, Pfaff, and Kunreuther 2011; Yin, Kunreuther, and White 2011). Whether actuarial pricing disciplines high-risk owners without simply driving them out of the market therefore remains unresolved.

Texas provides a clean natural experiment. In 1999 the state closed its public fund overnight and required all owners to purchase private coverage whose premiums vary with observable risk. I assemble a new facility-year panel covering 1990 to 2021 that links EPA inventories with administrative microdata from the Texas Commission on Environmental Quality and comparable data from eighteen control states that retained trust funds. The identification strategy rests on a make-model comparison sample that matches Texas facilities to control-state counterparts on the observable risk characteristics, tank age bin and wall construction type, that the private market prices. Because the treatment effect is theoretically heterogeneous along these pricing dimensions (Section 3, Proposition 3), comparability on them is required for valid causal inference; Section 4 shows that matching on make and model is what earns parallel pre-trends.

[REWIRTE RESULTS]

Related Literature

A well-established body of theoretical work suggests that strict liability rules can be an effective policy instrument for environmental risks, particularly when paired with insurance contracts that align a firm's private incentives with social costs (Shavell 1984c, 1984a, 1982,

2007; Polinsky 1980; Boyd and Kunreuther 1997). A crucial condition for this alignment is the insurer's ability to accurately price the risk of environmental harm. This paper provides the first empirical test of this principle in the context of environmental liability, offering causal estimates of how firms respond to a transition from flat-fee to more efficient, cost-based insurance contracts.

The challenge of pricing environmental risk connects this study to the economics of insurance rating regimes. The pooling equilibrium maintained by uniform-premium state funds is the environmental analog of community rating: premiums are invariant to observable risk, which creates cross-subsidies that suppress precautionary investment and crowd out risk-differentiated private alternatives (Rothschild and Stiglitz 1976; Pauly 1974; Cutler and Zeckhauser 2000). Theory predicts that a shift to experience rating generates two behavioral responses. On the selection margin, high-risk facilities face a cost increase and may exit or upgrade. On the moral hazard margin, facilities that remain now face a direct financial return to risk reduction, inducing investment in safer technology. Identifying these margins separately requires within-facility panel variation that aggregate data cannot provide. The empirical literature on rating regimes confirms both mechanisms, though rarely in the same market. In health insurance, risk adjustment reforms that moved Medicare Advantage toward experience rating reduced favorable selection but generated gaming responses among plans (Brown et al. 2014; McWilliams, Hsu, and Newhouse 2012). In credit markets, moving from flat-rate to risk-differentiated pricing improves screening and changes borrower composition (Einav, Jenkins, and Levin 2012, 2013; Nelson 2025), with efficiency gains depending on whether information asymmetries favor the insurer or the insured (Finkelstein and McGarry 2006; Liberman et al. 2018; Einav, Finkelstein, and Mahoney 2021). In the environmental insurance context, Yin, Kunreuther, and White (2011) find that switching from community-rated state funds to private coverage in Michigan reduced leak incidents by over twenty percent. The Texas reform studied here provides a sharper identification environment and facility-level data that allow the selection and moral hazard margins to be separated.

This paper also contributes to the literature on U.S. UST policy. While much of the existing work focuses on estimating the marginal damages of pollution from leaks (Marcus 2021; Zabel and Guignet 2012; Guignet 2014, 2013; Guignet and Martinez-Cruz 2018; Guignet et al. 2018; Walsh and Mui 2017), this study is complementary, examining the design of management policies intended to prevent those damages.

This research is most closely related to studies of a similar insurance market reform in Michigan by Yin, Pfaff, and Kunreuther (2011), Yin, Kunreuther, and White (2011), and Yin, Kunreuther, and White (2007). That body of work finds that transitioning to risk-based insurance reduced the number of small firms and reported leaks. However, those analyses relied on aggregate or cross-sectional data, which cannot control for unobserved facility-level heterogeneity and may be susceptible to omitted variable bias. This paper advances on prior work by adopting an empirical strategy similar to other modern studies of environmental liability (Boomhower 2019). By using a rich, facility-level panel dataset and a comparison sample matched on observable risk characteristics, I implement difference-in-differences models that control for unobserved facility characteristics and common time shocks, providing

more credible causal estimates of how firms adjust their technology, replacement, and exit decisions in response to cost-based pricing.

Institutional Background

Underground Storage Tanks: Physical Characteristics, Regulatory Lock-in, and Environmental Risk

The U.S. Environmental Protection Agency defines a UST system as any tank and connected underground piping with at least 10% of its volume below the surface (EPA 2024). The construction technology of a UST system is the primary determinant of its environmental risk profile and the observable margin on which actuarial pricing differentiates. The critical physical distinction within this capital stock is between single-walled systems (bare steel or fiberglass shells with no interstitial containment layer) and double-walled systems (which incorporate a secondary wall and continuous interstitial monitoring space).

The construction choice that creates this risk profile is economically consequential and effectively irreversible. Single-walled construction is cheaper to install: a 1987 GAO survey priced a standard 10,000-gallon single-walled unit at approximately $23,000 versus $39,000 for a comparable double-walled unit, a 67% premium that contemporaneous manufacturer filings confirm has persisted (GAO 1992; Containment Solutions 2021; Xerxes 2018). The savings are locked in at installation. Once a single-walled tank is in the ground, the owner cannot upgrade the wall construction without removing and replacing the tank entirely, a capital expenditure comparable to installing new equipment from scratch. The practical irreversibility of this choice is confirmed directly in the administrative data: in Texas from 1998 to 2021, only 0.003% of active facilities ever retrofitted tank construction prior to terminal closure. Wall type is therefore not a choice variable that responds to price signals at the intensive margin. It is a fixed, observable, and verifiable physical attribute, precisely the conditions required for an actuarially feasible pricing margin and the reason the private insurance market in Texas conditions premiums directly on it.

The regulatory framework reinforces this physical lock-in with a legal one. Under 40 CFR Part 280, UST systems are subject to vintage-differentiated technical standards: tanks installed under the 1988 regulations are grandfathered into the compliance classification they earned at the federal deadline, provided they remain unmodified. Any construction activity that alters the wall structure or piping configuration of an existing system, including the partial upgrades that would otherwise be the natural intermediate response to a rising insurance premium, triggers reclassification of the entire facility under the most current technical standards in effect at the time of modification. The 2005 and subsequent EPA technical revisions substantially raised those standards (US EPA 2021). An owner who initiates a partial upgrade to reduce premium costs therefore faces the prospect of a facility-wide compliance overhaul whose cost dwarfs the incremental premium savings.

This is a direct instance of the mechanism Gruenspecht (1982) identifies in the context of auto emissions: vintage-differentiated regulation that exempts existing capital from new standards inadvertently extends the operating life of the oldest and most hazardous stock by making any modification prohibitively costly. The practical effect here is to make the choice set binary: operate the existing system without modification, or close it entirely. As

Sallee (2023) argues more broadly, when new-capital regulation cannot reach the existing stock, retirement-focused instruments are the appropriate complement. The UST setting illustrates precisely why: incremental investment in risk reduction is not an option the regulatory environment makes available, which is why the extensive margin of closure is the sole behavioral margin the empirical analysis can and should target.

The environmental challenge posed by underground storage tanks is fundamentally a problem of managing an existing stock, not regulating new flows into it. As **?@fig-sw-share** documents, single-walled tanks constituted the majority of the active fleet at the federal compliance deadline and have declined only gradually over the study window, remaining a substantial share of active facilities through the 2010s. This stock is the source of the vast majority of confirmed releases: single-walled tanks exhibit a higher probability of a confirmed release than double-walled tanks at every age bracket, with the hazard gap widening after 15 years of operation (Section~**??**). New installations, by contrast, are overwhelmingly double-walled under post-1988 federal technical standards, so the marginal addition to the operating fleet poses little incremental risk. The remediation liability sitting beneath the surface today was installed decades ago and cannot be neutralized by tightening standards on new entrants. It can only be reduced by accelerating the exit of the single-walled incumbents. Because the hazardous technology is already in the ground, the relevant policy question is not how to screen entrants but how to price the continued operation of the existing fleet so that owners internalize the expected cost of a release.

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Notes: Share of active facilities with at least one single-walled tank in each calendar year, separately for Texas (orange) and the control-state pool (blue). The vertical dashed line marks the federal EPA upgrade deadline of December 22, 1998. The gradual post-deadline decline reflects attrition of the legacy single-walled fleet rather than any discrete regulatory shock, confirming that stock management rather than new-flow regulation governs the trajectory of environmental risk.

The Federal Regulatory Framework and UST Facilities Constrained Choice Set

Federal regulation of USTs originates in the 1984 Hazardous and Solid Waste Amendments (HSWA) to the Resource Conservation and Recovery Act, which directed the EPA to establish comprehensive technical and financial standards for tank systems. The EPA finalized these technical standards in 40 CFR Part 280 in 1988, requiring all non-compliant tanks, those lacking corrosion protection, spill and overfill controls, and release detection, to be upgraded, replaced, or closed by a uniform federal compliance deadline of December 22, 1998 (Agency 2017; US EPA 2021). In the years immediately preceding this deadline, the average annual facility-level closure rate spiked as the industry retired the oldest and most non-compliant portion of the capital stock. Following the deadline, the closure rate fell, producing a persistently aging legacy fleet whose slow turnover is an institutional feature of the stock the empirical analysis examines.

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Notes: Full sample closure rates for Texas and control-state facilities across the study window, illustrating the spike in closures driven by the federal EPA upgrade deadline of December 22, 1998, and the subsequent decline in annual exit rates as the surviving fleet aged.

Financial Responsibility and the State Trust Fund Regime

The 1988 Financial Responsibility Rule (40 CFR Part 280, Subpart H) mandates that every UST owner demonstrate the financial capacity to fund corrective action and satisfy third-party liability claims, with minimum required coverage of $1 million per occurrence (40 CFR §280, Subpart H). In the late 1980s, the nascent private environmental insurance market was unable to supply this coverage at affordable cost to the population of small, single-site operators that constitute the majority of UST owners. To prevent widespread non-compliance, many states established public trust funds financed by annual tank registration fees or fuel tax revenues, functioning as a publicly administered insurer of last resort. A defining feature of these funds is their premium structure: annual fees are set on a uniform, per-tank basis, invariant to any observable facility risk characteristic such as wall type, tank age, or leak detection technology. States retaining public funds consistently report that these funds cover 90% to 100% of all active facilities, as documented by the Association of State and Territorial Solid Waste Management Officials (ASTSWMO 1999--2024). This near-complete market saturation is the predicted equilibrium outcome: because the subsidized uniform-premium contract is priced below any risk-differentiated private alternative for high-risk facilities, private insurers are structurally unable to compete for the portion of the market with above-average risk. The resulting uniform-premium pooling equilibrium is the institutional baseline against which Texas's transition is evaluated, and it is precisely the community rating equilibrium that the insurance economics literature predicts will simultaneously generate adverse selection into the pool and suppress prevention investment at the facility level (Rothschild and Stiglitz 1976; Pauly 1974; Cutler and Zeckhauser 2000).

Texas's Policy Transition

Texas's departure from the public fund model constitutes a sharp, administratively determined policy discontinuity whose timing was driven by fiscal insolvency rather than any observable change in the statewide distribution of facility-level risk. Texas established the Petroleum Storage Tank Remediation Fund (PSTRF) in 1989 under the Texas Water Code. By 1995, actuarial assessments confirmed that the fund's liabilities, driven by the accumulating backlog of legacy remediation claims, were structurally unsustainable. In response, the 74th Texas Legislature enacted House Bill 2587, which mandated the sunset of the PSTRF (Texas HRO 1995). The legislature set the fund closure cutoff at December 22, 1998, the same calendar date as the federal EPA upgrade deadline codified in 40 CFR §280.21. The alignment of these two dates is not coincidental: the legislature explicitly selected the federal

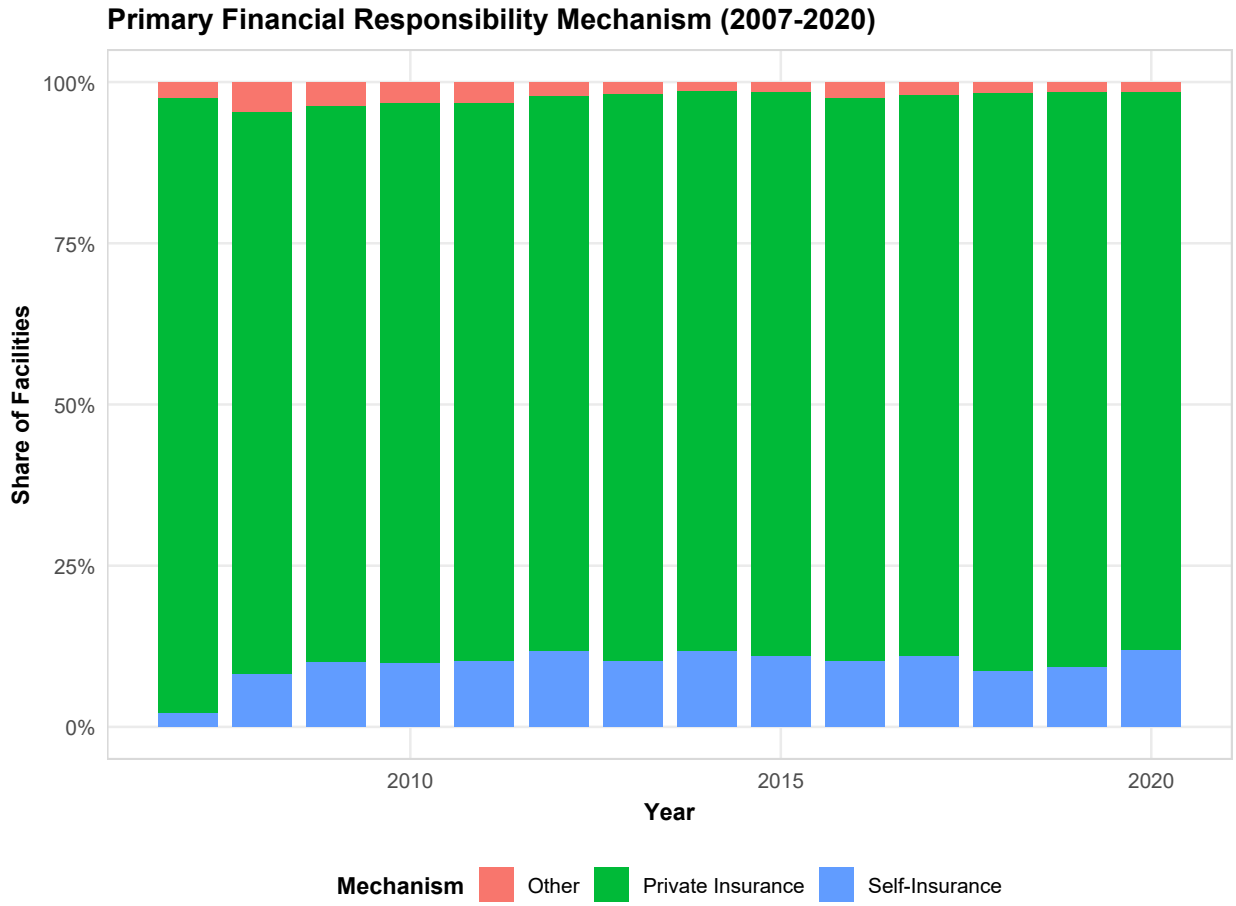
mandate date as the administrative anchor for the PSTRF cutoff, ensuring the fund would cease accepting new claims at precisely the moment when the largest cohort of non-compliant tanks was required to be addressed (74th Texas Legislature 1995). Effective December 22, 1998, all Texas UST owners were required to secure private insurance or qualify for self-insurance by demonstrating sufficient financial assets.

The Post-Transition Private Market in Texas

The private market that absorbed the Texas UST population after December 1998 prices observable facility risk directly, generating the cross-sectional and intertemporal premium variation that identifies firm behavioral responses. From 2007 to 2021, the period for which granular financial responsibility administrative data are available from the Texas Commission on Environmental Quality (TCEQ), an average of 89.8% of active retail UST facilities in Texas satisfied their financial responsibility obligation through commercial private insurance. The remaining 10.2% relied on self-insurance mechanisms, available exclusively to large, financially capitalized owners who can satisfy stringent asset tests.

Figure 1 documents the market transition directly. Financial responsibility coverage rates and the count of covered facilities confirm that adoption of private insurance was rapid and near-universal immediately following the December 1998 cutoff. The transition was not gradual: virtually the entire Texas UST population moved from public to private coverage within the first year of the reform.

Figure 1: Financial Responsibility Compliance in Texas, 2007–2021.



Notes: Left panel: share of active retail facilities satisfying the EPA financial responsibility requirement via private insurance versus self-insurance. Right panel: total count of covered facilities. The near-universal adoption of private coverage within the first policy year confirms the reform was absorbed rapidly across the facility population. Source: TCEQ financial responsibility administrative records.

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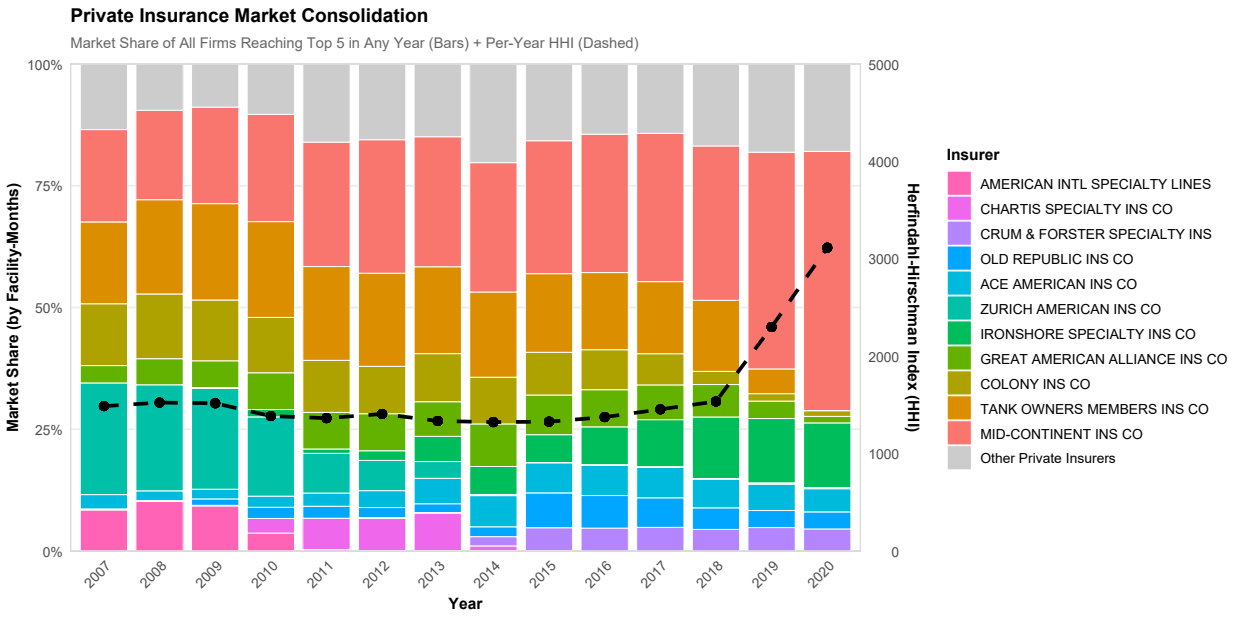
Figure 2: Market Concentration in the Texas UST Private Insurance Sector, 2007–2021.



Notes: Left panel: market share of the top five underwriters by covered facility count. Right panel: Herfindahl–Hirschman Index (HHI) for the Texas UST private insurance market. Mid-Continent Casualty Company (NAIC 23418) operated as the dominant underwriter throughout the post-transition period. Source: TCEQ financial responsibility administrative records.

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Private Insurance Premium Reconstruction

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Notes: Left panel: share of active retail facilities satisfying the EPA financial responsibility requirement via private insurance versus self-insurance. Right panel: total count of covered facilities. The near-universal adoption of private coverage within the first policy year confirms the reform was absorbed rapidly across the facility population. Source: TCEQ financial responsibility administrative records.

[THIS WHOLE SECTION NEEDS TO STATE CLEALRY THE FEW BEHAVIRAL ASSUMPTIONS I MADE TO RECREATE THE PREMIUM AS STATED IN THE CODE ABOUT DEDUCTABLE COICE AND THEN HOW THE BANDS ARE CREATED USING THE 40% CREDITS]

Private insurers do not publicly disclose individual policy premiums. I reconstruct the premium schedule that Texas facilities faced in the post-transition period by building a bottom-up actuarial rating engine from the rate filings of Mid-Continent Casualty Company (NAIC 23418). I obtain rate manuals filed with the System for Electronic Rates and Forms Filing (SERFF) covering four regulatory periods: 2006–2014, 2014–2019, 2019–2021, and 2021 onward. These filings provide the complete pricing algorithm, including base rates, Increased Limit Factors (ILF), and all secondary risk loads applied to observable facility characteristics.

I estimate the monthly premium P_{ijt} for tank j at facility i in month t as:

$$P_{ijt} = \text{Base}_t \times \text{ILF}_{it} \times \left(1 + \sum_k \lambda_{k,t} \mathbf{X}_{ijt}\right) \tag{1}$$

where Base_t is the per-tank base rate active in filing period t, ILF_{it} is the Increased Limit Factor adjusting for the elected coverage limit (defaulting to the statutory \$1 million per occurrence minimum where unspecified), and $\mathbf{X}_{ijt}$ is a vector of observable risk characteristics including tank age bin, wall construction type, piping type, and installed leak detection technology. The parameters $\lambda_{k,t}$ are the secondary risk loads, additive percentage credits or debits, assigned to each characteristic in filing period t.

Three implementation assumptions are required. First, I hold the base rate constant within each filing period at its filed value, consistent with the filing terms. Second, for risk characteristics unobserved in the administrative records (including aquifer proximity and flood zone classification), I set the corresponding risk load to zero, yielding a conservative estimate of the true premium. Third, the rate filings permit underwriters to apply schedule rating adjustments of \$±\$40% at their discretion; I use the base manual premium as the price signal and treat discretionary adjustments as mean-zero errors across the population, an assumption evaluated against the distribution of observed financial responsibility filings in the TCEQ records.

Granular premium microdata are available beginning in 2006. The period from January 1999 through December 2005 represents a transition window in which the private market was operational but individual premium records are not observed.

The Texas Phased Mandate: A Pre-Treatment Confound for the Oldest Cohort

Texas's transition to private financial responsibility was preceded by a separate regulatory regime that creates a pre-treatment confound for facilities holding pre-1988 tanks. Under 30 TAC Chapter 334, the Texas Natural Resource Conservation Commission implemented a staggered upgrade schedule for pre-1988 underground storage tanks, assigning compliance deadlines by tank vintage cohort that ran from December 22, 1989 through December 22, 1993 (Table 15). No analogous staggered state mandate existed in any of the control states

Table 1: Mandate Exposure Comparison: Texas Pre-1988 Cohort versus Control States.

[H]

Table 2: Mandate exposure comparison for pre-1988 tanks active December 22, 1998. Texas pre-1988 tanks cleared a cohort-specific compliance deadline 5–9 years prior; control state pre-1988 tanks faced their first federal deadline simultaneously with the insurance reform.

		Years Past Cohort Deadline as of Dec 22, 1998					
Group	N Tanks	Mean	Median	SD	Min	Max	% Past Deadline
Control	138748	0.0	0	0.0	0	0	0
Texas	47132	6.2	6	1.2	5	9	100

Years past deadline = (December 22, 1998) minus cohort mandate date. Texas tanks: mandate dates 1989–1993 by vintage cohort. Control tanks: mandate date = December 22, 1998 (0 years past).

during this period; control state pre-1988 facilities faced only the uniform federal December 22, 1998 deadline. This staggered schedule created a mechanical wave of compliance-induced closures among Texas pre-1988 facilities between 1989 and 1993 with no counterpart among control state facilities during the same years. It also fundamentally altered the composition of the surviving Texas pre-1988 tank stock by December 1998: the tanks still active at the insurance reform cutoff represent the subset whose operators had already decided, when confronted with a cohort-specific compliance deadline five to nine years earlier, to invest in upgrades rather than close. This survivor selection is documented in Table 1 and represents a direct threat to any naive comparison that pools pre- and post-1988 facilities without accounting for these differential pre-period dynamics. Section 4 documents how the make-model comparison sample addresses this and other sources of pre-treatment imbalance.

Notes: Comparison of cumulative closure rates and surviving fleet characteristics for Texas pre-1988 tank cohorts versus matched control-state counterparts. Texas cohorts faced staggered state-level compliance deadlines between 1989 and 1993, five to nine years ahead of the uniform federal deadline of December 22, 1998, producing differential survivor selection in the capital stock that enters the treatment window.

Control Group Definition and Institutional Restrictions

The control group comprises nineteen states that retained public uniform-premium trust funds continuously throughout the study window of 1985 to 2020, subject to three institutional complications that require explicit treatment.

[THIS SECTION NEEDS TO EXPLAIN WHY WE GO FROM THE 30ISH STATE FUND STATE TO THE 18/19 CONTROL STATES. USE THE HARMOINZED TANK DIAGNOSTICS TO EXPLAIN WHY: MAIN REASOSN ARE FROM THE EPA DATABASE MANY STATES ARE MISSING ONE OR MORE OF THE FOLLOWING DATA FOR ALL TANKS IN A STATE 1) INSTALL DATES OF AL LTANKS, CLOSURE DATES FOR TANKS FLAGGED AS CLOSED, POLLUTION REPORTED DATES FOR LUSTS OR UST CONSTRUCTION MATERIAL DEATILS TO CLASSIFY AS SINGLEW WALLED OR DOUBLE WALLED FACIITS.]

Insurance Regimes and the Retirement of Risky Capital

The insurance regime governs whether a facility's pollution risk enters its cost function. This section formalizes that claim, derives the closure threshold under each regime, and produces the testable predictions that motivate both the make-model comparison sample and the empirical specifications in Section 6. [MAKE THIS A SECTOIN REFF NOT HARD CODED]

Primitives

The facility owner's problem is a single-agent dynamic discrete choice model in the optimal stopping tradition (Rust, 1987). In an infinite-horizon setting, a competitive facility owner operates a single underground storage tank and each period decides whether to continue operating or permanently close. Let $\mathbf{x} \in \mathcal{X}$ denote the vector of facility characteristics that determine pollution risk: tank age, construction type, wall material, soil composition, and any other observable feature correlated with the probability of an environmental release. The instantaneous leak hazard is $h(\mathbf{x}) > 0$. For analytical tractability, I model $\mathbf{x}$ as the scalar tank age a, with $h'(a) > 0$ and $h''(a) \geq 0$. The qualitative results hold for any dimension of $\mathbf{x}$ along which the insurer conditions the premium. The Extension section below states this generalization explicitly.

Each leak event generates two categories of social harm. Remediation cost $L > 0$ is the direct cost of cleaning contaminated soil and groundwater. The operator bears this cost under strict liability, making it the natural object of an insurance contract. Health externality $E > 0$ represents diffuse health damages accruing to surrounding communities through groundwater contamination. Because contamination is frequently undetected for years after a release and attribution to a specific facility is administratively difficult, these damages generate no insurable third-party claim (Marcus (2021)). The gap between L and $L + E$ defines the limits of what any privately-priced insurance arrangement can achieve relative to the social optimum, and it motivates the welfare decomposition below.

The owner earns per-period net revenue $R > 0$ from operating and recovers scrap value $\kappa > 0$ upon permanent closure, representing the liquidity of the land net of decommissioning costs. The owner discounts at rate $\beta \in (0, 1)$ and is risk-neutral.

Insurance Regimes

The insurance regime determines whether and how the premium $C(\mathbf{x})$ varies with the facility's risk profile. I define three regimes that span the relevant policy space.

Definition 1 (Uniform Premium, UP). The facility pays a flat actuarially pooled premium each period, independent of any facility characteristic:

$$C_{\text{UP}}(\mathbf{x}) = \bar{p}.$$

No risk information is transmitted to the firm through the insurance contract under UP. In the scalar age case, the marginal cost of aging, the elasticity of the premium with respect to capital depreciation, is identically zero:

$$\frac{\partial C_{\text{UP}}}{\partial a} = 0.$$

The insurer may possess full information about the cross-sectional risk distribution but does not act on it in setting individual premiums. This is the pricing structure of state-administered petroleum storage tank assurance funds, including the Texas PST Fund prior to its 1999 sunset.

Definition 2 (Risk-Based Pricing, RB). The insurer prices actuarially, conditioning on all observable elements of $\mathbf{x}$. The premium equals expected claims scaled by a loading factor $\lambda \geq 1$:

$$C_{\mathrm{RB}}(a) = \lambda \cdot h(a) \cdot L.$$

Since $h'(a) > 0$, the premium's elasticity with respect to capital depreciation is strictly positive:

$$\frac{\partial C_{\mathrm{RB}}}{\partial a} = \lambda h'(a) L > 0.$$

Every dimension of $\mathbf{x}$ that enters $h(\mathbf{x})$ is mechanically injected into the firm's flow cost by the actuarial pricing rule. This is a definitional property of risk-based pricing, not a behavioral assumption about firm awareness or optimization.

Definition 3 (Social Optimum, SOC). The social planner internalizes both remediation costs and health externalities:

$$C_{\mathrm{SOC}}(a) = h(a) \cdot (L + E).$$

The social optimum is unattainable under private insurance because E generates no insurable claim. For the parametrically relevant condition $E > (\lambda - 1)L$, the cost ordering is:

$$C_{\mathrm{UP}} \leq C_{\mathrm{RB}}(a) < C_{\mathrm{SOC}}(a),$$

where the left inequality becomes strict for a large enough that $\lambda h(a) L > \bar{p}$.

The Firm's Dynamic Problem

Each period, the owner observes age a and chooses to maintain or permanently close. Under regime j, flow utility from maintaining is $u_j(a) = R - C_j(a)$, where R is annual net revenue from operations net of all non-insurance costs and $C_j(a)$ is the insurance premium. I maintain the simplifying assumption that insurance contracts provide full ex-post coverage with no deductible, treating environmental risk solely as an ex-ante premium cost rather than an ex-post liability shock. This assumption isolates the cost-of-continuing channel and is consistent with standard provisions in the petroleum storage tank insurance market; the structural estimation in Section 8 relaxes it to incorporate partial risk internalization through deductibles.

The Bellman equation under regime j is:

$$V_j(a) = \max \left\{ \underbrace{R - C_j(a) + \beta\, \mathbb{E}[V_j(a') \mid a]}_{\text{Maintain}}, \quad \underbrace{\kappa}_{\text{Close}} \right\},$$

where β is the discount factor, the expectation is taken over the stochastic evolution of tank age, and κ is the scrap value recovered upon exit. The pure continuation value, the present discounted value of flow utility under the assumption that the tank is never closed, is:

$$\widetilde{V}_j(a) = \sum_{s=0}^{\infty} \beta^s \left[R - C_j(a+s) \right].$$

The optimal value function satisfies $V_j(a) = \max\{\widetilde{V}_j(a),\, \kappa\}$. The optimal closure threshold is:

$$a_j^* = \min\left\{a \geq 0 : \widetilde{V}_j(a) \leq \kappa\right\}.$$

Under any regime with $C_j'(a) > 0$, the function $\widetilde{V}_j(a)$ is strictly decreasing, so the optimal policy is a threshold rule: close if and only if $a \geq a_j^*$.

Main Results

Proposition 1 (Risk Transmission and Value Slope). *Under UP, $\widetilde{V}_{UP}'(a) = 0$. Under RB, $\widetilde{V}_{RB}'(a) < 0$. Under SOC, $\widetilde{V}_{SOC}'(a) < \widetilde{V}_{RB}'(a) < 0$.*

Proof. Differentiating the continuation value with respect to age:

$$\widetilde{V}_j'(a) = -\sum_{s=0}^{\infty} \beta^s C_j'(a+s).$$

Under UP, $C_{\text{UP}}'(\cdot) = 0$ everywhere, so the sum is zero and the continuation value is flat in age: the physical degradation of the asset is real but generates no financial pressure on the firm. Under RB, $C_{\text{RB}}'(a) = \lambda h'(a)L > 0$, so every term in the sum is strictly negative and $\widetilde{V}_{\text{RB}}'(a) < 0$. Under SOC, $C_{\text{SOC}}'(a) = h'(a)(L+E) > \lambda h'(a)L = C_{\text{RB}}'(a)$, so the SOC slope is strictly more negative at every age. □

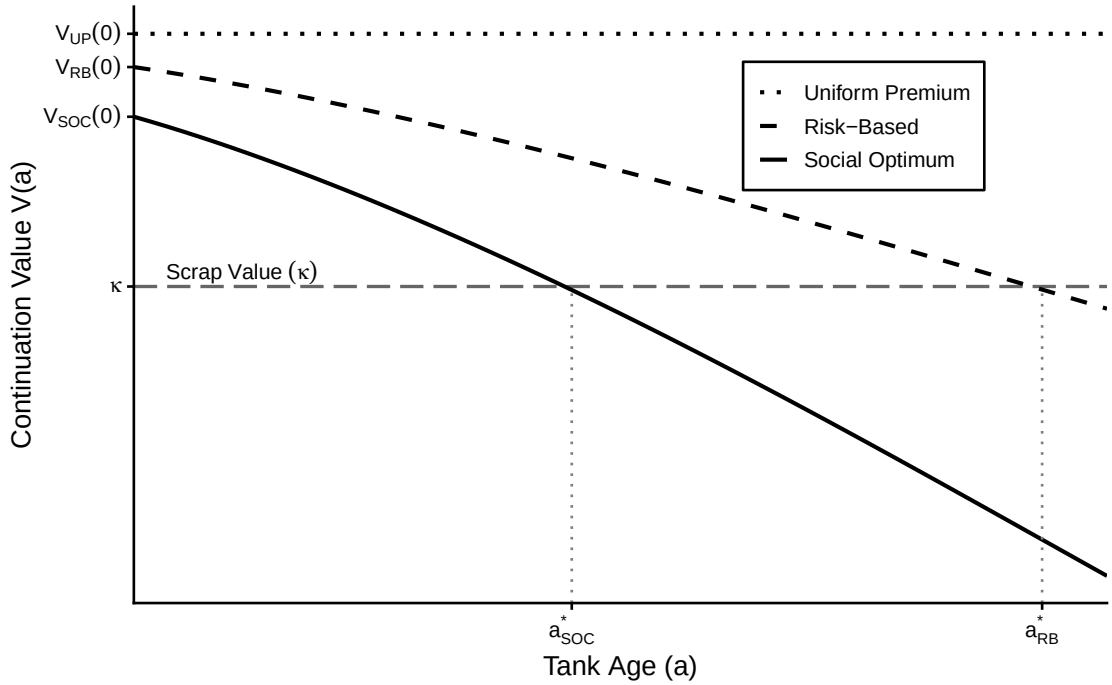
Proposition 1 establishes the mapping from the pricing regime to the information environment of the firm. Under UP, aging is economically invisible: the zero elasticity of the premium with respect to capital depreciation means the continuation value contains no signal about the facility's pollution hazard, and the firm retains the asset until it is physically obsolete. Under RB, aging is economically visible by construction: the actuarial pricing rule translates the insurer's risk model into a per-period cost that rises with the facility's hazard profile, pivoting the continuation value downward for older tanks and accelerating the optimal exit age, regardless of whether the owner attends to or understands the actuarial logic.

Proposition 2 (Regime Ordering of Closure Thresholds). *Under the condition $E > (\lambda - 1)L$, the optimal closure ages satisfy $a_{SOC}^* < a_{RB}^* < a_{UP}^*$.*

Proof. By Proposition 1, $\widetilde{V}_{\text{SOC}}(a) < \widetilde{V}_{\text{RB}}(a) < \widetilde{V}_{\text{UP}}(a)$ for all $a > 0$. Since a_j^* is defined as the first age at which $\widetilde{V}_j(a)$ crosses κ from above, a lower value function crosses the threshold at a younger age. The ordering of value functions is preserved under discounted summation because $h(a)$ is common across regimes; only the coefficient multiplying $h(a)$ differs. □

Figure 3 illustrates Proposition 2. The SOC curve declines at the fastest rate and crosses the scrap value line κ at a_{SOC}^*. The RB curve declines at an intermediate rate and crosses κ at $a_{\text{RB}}^* > a_{\text{SOC}}^*$, strictly reducing the optimal tank lifespan relative to the uniform pricing baseline. The UP curve is flat by Proposition 1 and does not cross κ within the plotted range, reflecting the zero elasticity of the premium with respect to capital depreciation under pooled pricing.

Figure 3: The Mechanism: Optimal Stopping under Alternative Insurance Regimes



Notes: This figure illustrates the optimal closure thresholds under the three insurance regimes derived in Proposition 2. The horizontal line represents the scrap value (κ). The Social Optimum (SOC) continuation value declines at the fastest rate, prompting the earliest closure at a^*_{SOC}. The Risk-Based (RB) pricing continuation value declines at an intermediate rate, yielding closure at a^*_{RB}. The Uniform Premium (UP) continuation value remains flat, reflecting the zero elasticity of the premium with respect to capital depreciation under pooled pricing.

Frictions Attenuating Behavioral Response

The model establishes the direction of the effect of moving from UP to RB pricing. It does not establish the magnitude, and it is silent on two frictions that could attenuate behavioral response even when the directional incentive is correctly signed.

The first friction is premium insufficiency. Even if $C'_{\text{RB}}(a) > 0$, the premium increment at the margin may be small relative to the fixed transaction costs of closure, which include demolition, regulatory notification, site remediation, and foregone operating revenue. If the present value of avoided future premiums is less than these fixed exit costs for the marginal tank, the regime shift generates no behavioral response despite correctly pricing risk.

The second friction is constraints on action. Even when $\widetilde{V}_{\text{RB}}(a) < \kappa$ and closure is optimal, the owner may be unable to act on that signal. Credit constraints may prevent financing upfront exit costs. Thin markets for contaminated properties may render the book value of κ unrealizable. Present bias or inattention to the actuarial premium schedule could cause owners to underweight future premium increases.

Neither friction is resolvable from theory alone. The Texas natural experiment provides the empirical test: the 1999 sunset of the PST Fund was abrupt, mandatory, and economically large, providing the variation

required to estimate whether the premium-burden mechanism is sufficient to produce measurable changes in closure behavior.

Proposition 3 (Orthogonality Under UP; Correlation Under RB). *Let ξ denote closure drivers unrelated to the facility's pollution hazard, land value appreciation, local fuel demand shocks, and owner liquidity or retirement decisions, satisfying $\xi \perp a$. Then: (i) under UP,* $\mathrm{Cov}(\mathbf{1}[\mathrm{Close}],\, a) = 0$*; (ii) under RB,* $\mathrm{Cov}(\mathbf{1}[\mathrm{Close}],\, a) > 0$*, and the treatment effect* $\Delta(a) \equiv \Pr(\mathrm{Close} \mid \mathrm{RB}, a) - \Pr(\mathrm{Close} \mid \mathrm{UP}, a)$ *is zero for $a < a^*_{RB}$ and strictly increasing in a for $a \geq a^*_{RB}$.*

Proof. Under UP the only closure drivers are elements of ξ, which satisfy $\xi \perp a$ by assumption, so $\mathrm{Cov}(\mathbf{1}[\mathrm{Close}], a) = 0$. Under RB, $\widetilde{V}_{\mathrm{RB}}(a)$ is strictly decreasing, so $\Pr(\widetilde{V}_{\mathrm{RB}}(a) \leq \kappa)$ is strictly increasing in a and the covariance is positive. For the treatment effect: when $a < a^*_{\mathrm{RB}}$, $\widetilde{V}_{\mathrm{RB}}(a) > \kappa$ and $\widetilde{V}_{\mathrm{UP}}(a) > \kappa$ under both regimes, so $\Delta(a) \approx 0$. When $a \geq a^*_{\mathrm{RB}}$, $\widetilde{V}_{\mathrm{RB}}(a) \leq \kappa < \widetilde{V}_{\mathrm{UP}}(a)$, so the regime difference is the binding margin and $\Delta(a) > 0$, strictly increasing as the gap $\widetilde{V}_{\mathrm{UP}}(a) - \widetilde{V}_{\mathrm{RB}}(a)$ widens with age. □

The $\xi \perp a$ assumption warrants comment. It requires that non-insurance closure drivers, including owner retirement, land value appreciation, and local demand shifts, are mean-independent of tank age. This is an identifying assumption, not a definitional one. If owners of older tanks are themselves older and more likely to retire, or if facilities in older commercial districts face systematically different land value trajectories, the orthogonality would be violated. The pre-period evidence in Section 4.5 provides an indirect test: under UP, control-state closure rates should exhibit no age gradient in the pre-treatment period. The parallel trends evidence is consistent with this condition, though it cannot rule out violations that are small relative to sampling noise.

Proposition 3 is the identifying prediction of the paper. The shift from UP to RB does not merely change the number of tanks that close; it reorganizes *which* tanks close. Under UP, the age distribution of closures is determined entirely by land value appreciation, demand contraction, and owner liquidity or retirement, drivers that are mean-independent of the facility's hazard profile. Under RB, the age distribution of closures becomes correlated with the hazard profile in a specific, premium-gradient-shaped way: the treatment effect is zero for young tanks and rising for old tanks, with the crossover at a^*_{RB}.

This heterogeneous treatment effect has a direct implication for identification. If the treatment effect is concentrated among older, higher-risk facilities, a comparison group that is unbalanced on age or wall type will produce biased estimates, not because the treated and control groups follow different trends on average, but because the composition of each group puts different weight on the margin where the treatment bites. The make-model comparison sample, introduced in Section 4, is constructed precisely to address this: by matching on age bin and wall type, the comparison group is balanced on the dimensions along which $\Delta(a)$ varies, removing the compositional source of bias.

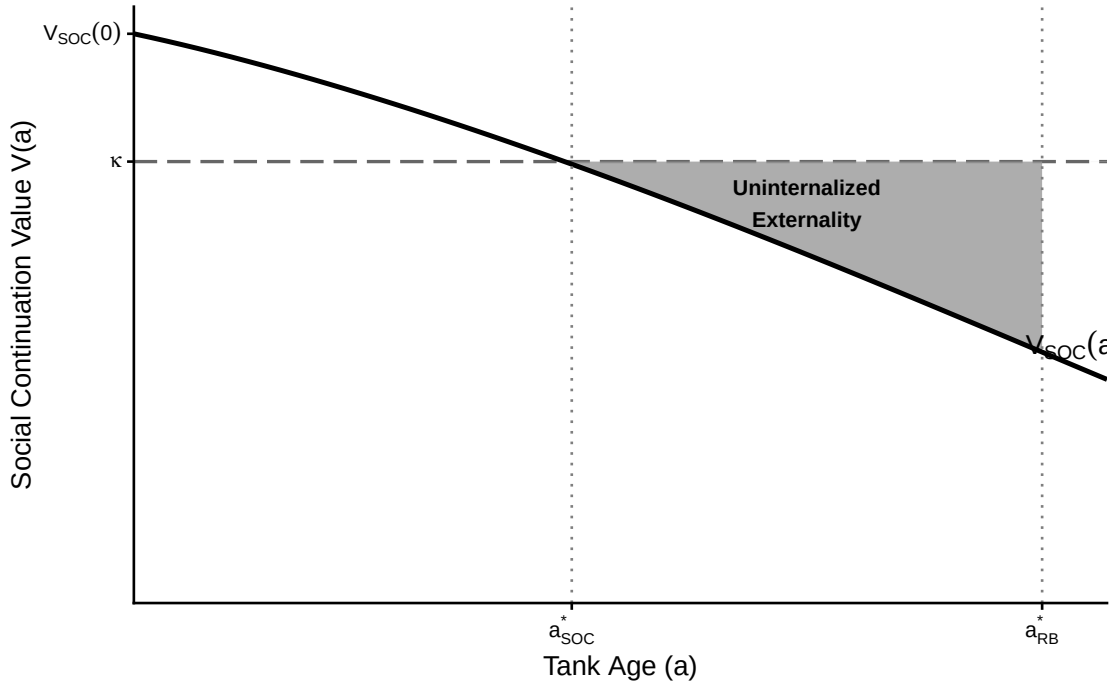
Welfare Decomposition

The welfare cost of retaining a tank beyond the socially optimal closure age is the expected externality accumulated during the excess operating period. The total welfare loss from UP decomposes into two components:

$$\mathrm{DWL}_{\mathrm{UP}} = \underbrace{\int_{a^*_{\mathrm{SOC}}}^{a^*_{\mathrm{RB}}} h(a)\, E\, da}_{\text{Unavoidable Wedge}} + \underbrace{\int_{a^*_{\mathrm{RB}}}^{a^*_{\mathrm{UP}}} h(a)\, E\, da}_{\text{Eliminable Excess}}.$$

The unavoidable wedge exists because E generates no insurable claim, so the gap $a^*_{\mathrm{RB}} - a^*_{\mathrm{SOC}}$ cannot be closed by insurance design alone. The eliminable excess is the welfare loss attributable specifically to flat-fee pooling, the component recoverable by moving from UP to RB pricing.

Figure 4: Welfare Analysis: The Externality Wedge and Partial Correction



Extension to the Full Risk Vector

The model uses age as a scalar proxy for $\mathbf{x}$ for tractability. Propositions 1 through 3 hold for any dimension of $\mathbf{x}$ along which (i) the hazard $h(\mathbf{x})$ varies and (ii) the insurer conditions the premium. The mechanism is identical: any risk factor that enters the actuarial pricing schedule is mechanically injected into the firm's flow cost, raises the elasticity of the premium with respect to that risk dimension, and generates closure sensitivity that is absent under flat-fee pooling.

This extension has a concrete empirical counterpart in the petroleum storage tank market. Private insurers in Texas condition premiums on both age and tank wall construction type, with single-wall tanks facing premiums approximately twice those of double-wall tanks of equivalent age, as documented in Mid-Continent rate filings for the 2007 to 2021 period. If the mechanism of Proposition 3 extends to all priced risk dimensions, closure decisions under RB pricing should become correlated with wall type in the same way they become correlated with age. I treat this as an empirical generalizability test: a positive result for wall type corroborates the premium-burden channel; a null result would indicate either that wall-type premium gradients are insufficient to move behavior or that single-wall operators face structural exit barriers distinct from those facing aged-tank operators.

Testable Predictions

The following four hypotheses map directly from the propositions above to the difference-in-differences specifications in Section 6. In each case, the empirical test compares Texas facilities, which transitioned from UP to RB pricing in 1999, to make-model-matched control-state facilities, which retained flat-fee public insurance programs throughout the sample period.

Hypothesis 1 (Age Sensitivity of Closure). Under UP, the closure hazard is mean-independent of a (Proposition 3, part i). Under RB, the positive elasticity of the premium with respect to capital depreciation

creates a premium-burden channel that makes older facilities more likely to close (Proposition 3, part ii). The prediction is that the age gradient in the closure hazard steepened differentially in Texas after 1999:

$$\frac{\partial \lambda(t \mid a, \mathrm{RB})}{\partial a} - \frac{\partial \lambda(t \mid a, \mathrm{UP})}{\partial a} > 0.$$

A discrete-time hazard model with the triple interaction TX $\times$ Post $\times$ a is the natural estimator. The linear probability model on the binary closure indicator is a valid approximation under the assumption that censoring is independent of age conditional on covariates; the survival model relaxes this assumption.

Hypothesis 2 (Heterogeneous Treatment Effect by Age). The treatment effect of switching from UP to RB is zero for young tanks and strictly positive and increasing for old tanks (Proposition 3). For cutoff age $\bar{a}$:

$$\mathrm{ATT}(a > \bar{a}) > \mathrm{ATT}(a \leq \bar{a}) \approx 0.$$

Hypothesis 3 (Rightward Shift in Age Distribution of Closures). The transition to RB pricing triggers a concentrated wave of exits among older tanks, raising the expected age of a closing facility relative to control-state counterparts:

$$\mathbb{E}[a \mid \text{Close, Texas, Post}] > \mathbb{E}[a \mid \text{Close, Control, Post}].$$

The mean shift is the testable corollary of a richer distributional prediction: under RB, the age-at-closure distribution concentrates mass at high ages near a^*_{RB}, whereas under UP the distribution of age at closure mirrors the age distribution of the full operating population. A test of the mean shift conditions on the selected sample of closures, which introduces selection bias because the tanks that close under RB are a non-random subset of the population. The density of age at closure by group and period provides the non-parametric visual companion to the regression test.

Hypothesis 4 (Wall-Type Sensitivity). If the mechanism extends to all priced dimensions of $\mathbf{x}$, the single-wall versus double-wall gap in the closure hazard should widen under RB pricing relative to UP:

$$[\lambda(t \mid \text{SW, RB}) - \lambda(t \mid \text{DW, RB})] > [\lambda(t \mid \text{SW, UP}) - \lambda(t \mid \text{DW, UP})]\,.$$

The empirical test is the triple interaction TX $\times$ Post $\times$ SingleWall in the closure specification.

Three empirical patterns would be inconsistent with the premium-burden channel and would instead indicate confounding from a Texas-specific economic shock: a uniform treatment effect across all age and wall-type groups; a closure response concentrated among young or low-risk facilities; or no correlation between the treatment effect magnitude and observable risk factors. These patterns define the falsification criteria for the identification strategy in Section 6.

Table 3: Sample Attrition Across Five Sequential Construction Filters.

[!h]

Table 4: Sequential attrition log. Each row applies one additional filter to the facility-year panel. Δ columns show marginal change in observations.

Step	Stage	Filter	Facilities	Fac-Years	Δ Facilities	Δ Fac-Years
0	Raw harmonized panel	All states, all years	478503	12702871	NA	NA
1	Study window	[1985, 2020]	469681	7985050	-8822	-4717821
2	Analysis states	TX + 19 controls (NJ excl.)	306731	4978288	-162950	-3006762
3	Incumbent sample	first_observed < 1998	286882	4705725	-19849	-272563
4	Complete records	Facility-level date exclusion (53,648 excl.)	233234	3967583	-53648	-738142
5	Final analysis sample		233234	3967583	0	0

Data

Panel Architecture and Source Layers

The analysis uses a facility-year panel built by merging three administrative data sources. The merge key is a panel identifier constructed from the state-standardized facility code and the two-letter state abbreviation. The first source is the behavioral backbone: a facility-year panel compiled from state UST administrative inventories and EPA summary files covering Texas and the nineteen control states from 1985 through 2020. This layer records, for each facility in each year, whether the facility closed a tank, permanently exited the market, retrofitted its tank stock, or reported a confirmed leak. The second source is a tank-level physical inventory recording the installation date, closure date, wall construction type, capacity, fuel type, and operational status for every tank associated with each facility. The third source is financial: private insurance premium estimates for Texas facilities reconstructed from actuarial rate filings, and realized cleanup cost records drawn from state trust fund payment ledgers for a subset of control states. The unit of observation throughout is the facility-year.

Sample Construction and Attrition

The estimation sample is built through five sequential filters, each motivated by a specific identification requirement. Table 3 reports the facility count and facility-year count remaining after each filter.

The first filter restricts the panel to the 1985 through 2020 study window, the period over which administrative data coverage is sufficiently continuous across treated and control states to support pre-trend testing and post-treatment tracking. The second filter restricts the sample to Texas and the nineteen control states defined in Section 2.7. The third filter keeps only incumbent facilities, defined as facilities first observed in the administrative records before the 1998 statutory treatment year. This restriction ensures that every facility in the sample faced the reform as an unanticipated constraint rather than an entry condition, which eliminates compositional changes in the market following the reform announcement.

The fourth filter drops facilities with missing tank-level date records. A facility is excluded if any of its tanks lacks a recorded installation date, or if any tank recorded as closed lacks a corresponding closure date. Table 5 reports the balance test for this exclusion. A logistic regression of the missing-date indicator on the Texas treatment dummy yields a statistically insignificant coefficient ($p = 0.5361$), confirming that the probability of exclusion is unrelated to treatment status. The fifth filter applies the make-model common support restriction developed in the following section, which retains only facilities whose wall type and age profile falls within the range of characteristics observed in both Texas and the control group as of December 22, 1998.

Table 5: Balance Test for the Missing-Data Exclusion.

[!h]

Table 6: Balance test for missing date exclusion. Logistic regression of *any missing date* indicator on Texas dummy. $p < 0.05$ would indicate differential exclusion.

term	estimate	std.error	statistic	p.value
(Intercept)	-1.2061	0.0049	-245.0630	0.0000
texas	-17.3600	28.0601	-0.6187	0.5361

Notes: Each row reports the facility count and facility-year count remaining after applying cumulative filters, separately for Texas and the control group. Filters are applied sequentially: (1) study window restriction, 1985–2020; (2) Texas and nineteen control states; (3) incumbent facilities only, first observed prior to 1998; (4) exclusion of facilities with missing tank-level installation or closure dates; (5) make-model common support restriction on portfolio wall type and oldest-tank age bin as of December 22, 1998.

Notes: The dependent variable equals one if the facility was excluded from the estimation sample due to missing tank-level date records. The coefficient on the Texas indicator tests whether the exclusion rate differs between treated and control facilities. A statistically insignificant coefficient ($p = 0.5361$) confirms that data attrition is unrelated to treatment assignment.

Primary Outcome Variables

I construct three binary outcome variables at the facility-year level from the behavioral backbone. The primary outcome, Closure_{it}, equals one if facility i recorded at least one tank closure in calendar year t. Among facility-years where $\text{Closure}_{it} = 1$, I construct two mutually exclusive conditional outcomes: Exit | Closure_{it}, equal to one if the facility permanently ceased operations in year t, and Replace | Closure_{it}, equal to one if the facility closed a tank but continued operating, implying on-site replacement. The environmental outcome, Leak_{it}, equals one if facility i reported a confirmed release to state regulators in year t, drawn from the harmonized Leaking Underground Storage Tank (LUST) incident records. All outcomes are defined at the facility-year level by aggregating underlying monthly records to an annual frequency.

For the event study specifications, I define relative event time as $k = \text{year} - 1999$, indexed to the first full post-treatment calendar year, so that $k = -1$ corresponds to calendar year 1998, the omitted baseline period. The event study window spans 1985 to 2018.

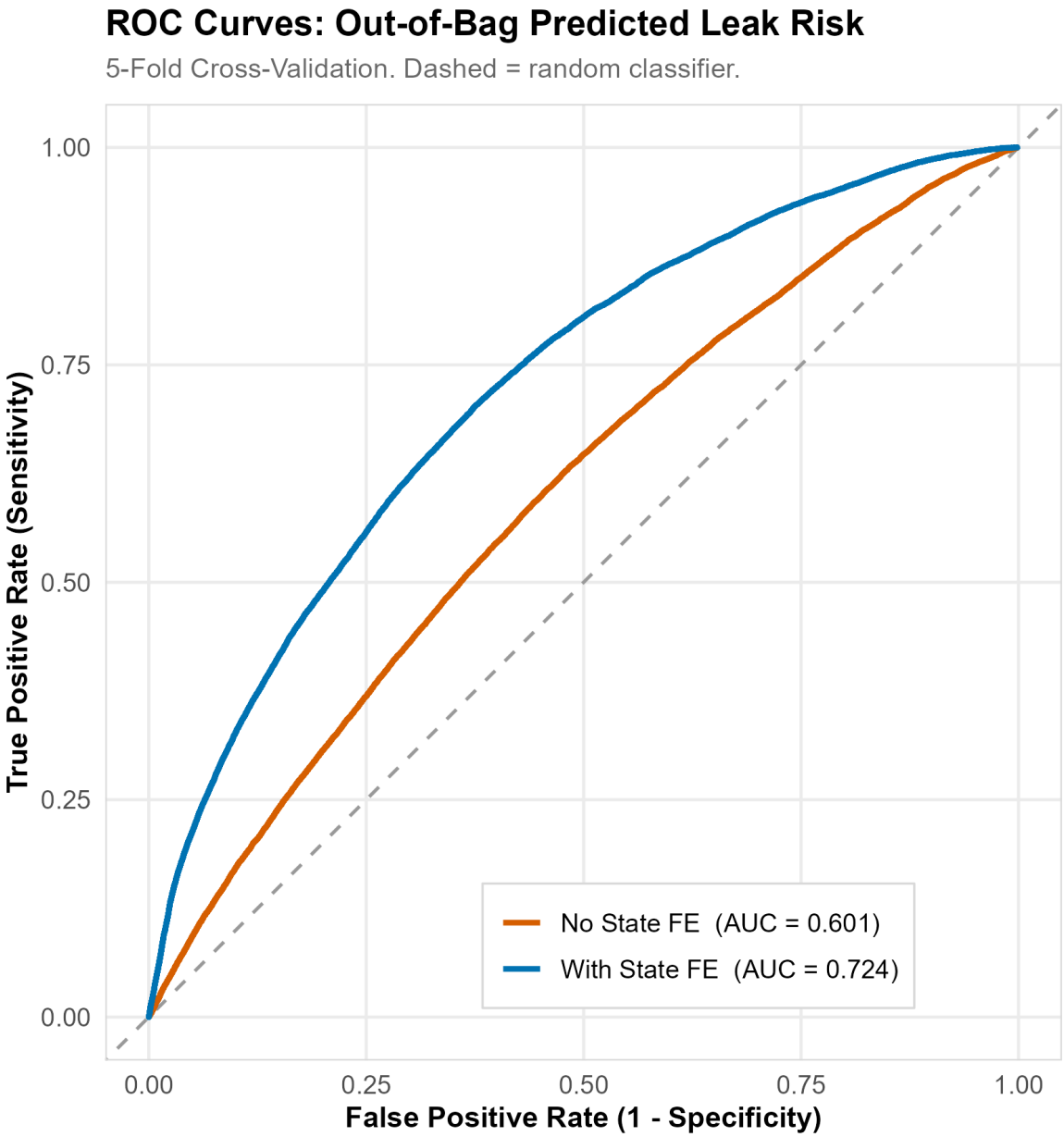
Observable Risk Characteristics and the Actuarial Pricing Foundation

Risk-based insurance can only discipline facility operators if the characteristics private insurers observe actually predict future environmental liability. If tank age and wall construction type were unrelated to release probability and cleanup cost, insurers would have no basis for charging different premiums across facilities, and the premium gradients documented in Section 2.5 would be noise rather than signal. This section shows that the two characteristics at the center of the theoretical mechanism do in fact predict both confirmed releases and realized remediation costs, and that this predictability holds out-of-sample. This establishes that the premium schedule documented in the rate filings reflects genuine actuarial content, and it motivates the comparison sample construction in the following section.

Release Probability Is Predictable from Observables

Age bin and wall construction type predict confirmed releases out-of-sample. Figure 5 presents receiver operating characteristic curves from cross-validated models that predict whether a facility reports a confirmed release in a given year, using only age bin and wall type as inputs. The models are estimated on pre-treatment control-state facility-years and evaluated on held-out data. The area under the curve exceeds the no-skill baseline in both the specification without state fixed effects (left panel) and the specification with state fixed effects (right panel). Observable facility characteristics therefore carry genuine predictive content for releases above and beyond cross-state differences in baseline release rates, and private insurers have an actuarially sound basis for premium differentiation.

Figure 5: Cross-Validated ROC Curves for Predicting Confirmed Releases from Observable Facility Characteristics.



Notes: Predictors are age bin and wall construction type. Left panel: model without state fixed effects. Right panel: model with state fixed effects. The area under the curve exceeds the no-skill baseline (dashed diagonal) in both specifications. Estimated on pre-treatment control-state facility-years. Source: 01n.

?@fig-pdp-age-wall shows the direction and shape of these risk gradients. The predicted release probability rises with age bin for both single-wall and double-wall tanks, consistent with the model assumption $h'(a) > 0$. The gap between single-wall and double-wall predicted probabilities also widens with age: the single-wall penalty is not a constant offset but grows as tanks age. These gradients are estimated from pre-treatment

control-state data, not imposed by the model, and they match the structural form of the premium schedule in the Mid-Continent rate filings.

`Figure file not found at: C:/Users/kaleb/Documents/ust_ins_move_to_github/Output/Figures/Figure_CV_Part`

Notes: Separately for single-wall (solid) and double-wall (dashed) tanks, with observed facility-year means overlaid. The predicted hazard rises with age for both wall types, and the single-wall penalty widens with age. Source: 01n.

The predicted risk scores are also well-calibrated against observed release rates. **?@fig-calibration** plots mean observed release rates against mean predicted scores within deciles of the score distribution. The data points fall close to the 45-degree line across the full risk range, confirming that the model produces reliable probability estimates and not just a ranking. This calibration supports the use of the predicted scores as a continuous risk measure in the structural model of Section 8.

`Figure file not found at: C:/Users/kaleb/Documents/ust_ins_move_to_github/Output/Figures/Figure_CV_Cali`

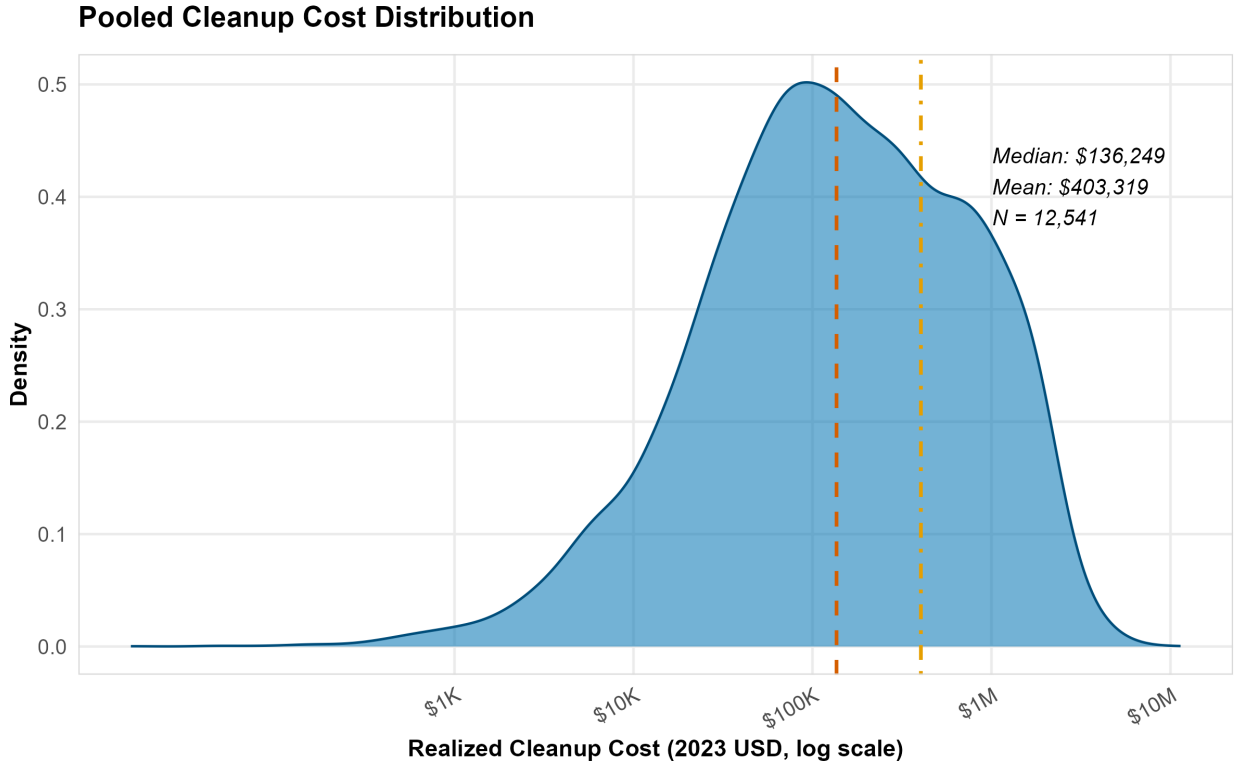
Notes: Mean observed release rate against mean predicted risk score within deciles of the predicted score distribution. Points near the 45-degree line indicate a well-calibrated model. Source: 01n.

Remediation Costs Also Scale with Observable Risk

For risk-based pricing to produce the welfare improvement identified in Section 3, premiums must reflect expected damages, which is the product of release probability and conditional cleanup cost. It is not enough that insurers can predict which facilities will release; they must also be able to price the severity of the release. I establish that remediation costs follow the same observable gradients as release probability, using payment ledger data from state underground storage tank trust funds in Colorado, Louisiana, New Mexico, Pennsylvania, Tennessee, and Utah. Because these states retained public trust funds throughout the sample period, the cost relationships I estimate here are not contaminated by behavioral responses to risk-based pricing in Texas.

Figure 6 shows the pooled log-scale distribution of realized cleanup costs across the six states, deflated to 2023 dollars. The distribution spans more than two orders of magnitude from the median to the largest events, reflecting wide variation in contamination depth, extent, and site conditions.

Figure 6: Pooled Distribution of Realized Cleanup Costs per Confirmed Release.



Notes: Across six state trust funds (Colorado, Louisiana, New Mexico, Pennsylvania, Tennessee, Utah), deflated to 2023 dollars. Log-scale density with vertical lines at the median (dashed red) and mean (dot-dash orange).

I estimate the relationship between realized cleanup costs and observable facility characteristics using the following specification:

$$\log(\text{Cost}_{it}) = \sum_{b} \alpha_b \cdot \mathbf{1}[\text{AgeBin}_{it} = b] + \mathbf{X}'_{it}\gamma + \mu_s + \delta_t + \varepsilon_{it} \tag{2}$$

where Cost_{it} is total realized remediation expenditure for facility i in claim-year t, deflated to 2023 dollars; $\mathbf{1}[\text{AgeBin}_{it} = b]$ are indicators for the facility's mean tank age bin at the time of the confirmed release, with the youngest bin (0–5 years) as the omitted category; $\mathbf{X}_{it}$ is a vector of facility-level controls comprising active tank count, total tank capacity in thousands of gallons, and the facility's single-wall share measured as a tercile indicator (0–33% as baseline, 34–66%, 67–100%, and an unknown category absorbing New Mexico facilities where wall type is unrecorded); μ_s are state fixed effects; δ_t are year fixed effects; and ε_{it} is an idiosyncratic error term. Standard errors are clustered at the state level. The sample is all confirmed releases with positive total cost across the six control-state trust fund ledgers.

Table 7 reports the results across four specifications. Column (1) regresses log cost on the age bin indicators alone. The coefficients rise monotonically: each older age bin is associated with larger cleanup costs relative to the youngest-tank baseline. Column (2) adds the facility-level controls. The single-wall share tercile indicators show a dose-response pattern, with facilities in the highest single-wall tercile facing the largest

cost penalty. Column (3) adds state fixed effects and Column (4) adds year fixed effects. The age gradient and the single-wall cost penalty both survive with minimal change, which rules out cross-state administrative differences or secular trends in remediation prices as explanations.

Notes: Dependent variable: log realized cleanup cost (2023 USD), per equation~2. Column (1): age bin indicators only. Column (2): adds active tank count, total capacity, and single-wall share tercile. Column (3): adds state fixed effects. Column (4): adds state and year fixed effects. Standard errors clustered at the state level. Sample: confirmed releases with positive total cost across six control-state trust funds (CO, LA, NM, PA, TN, UT).

Table 7: Descriptive Correlates of Realized Cleanup Costs.

Dependent Variable:	log(Cleanup Cost, 2023 USD)			
	Bivariate	Controls	State FE	State+Year FE
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	12.06***	11.26***		
	(0.1349)	(0.0569)		
age_bins5-10	0.0853	-0.1541*	-0.2100***	-0.0761*
	(0.0441)	(0.0612)	(0.0308)	(0.0351)
age_bins10-15	-0.1612	-0.4734***	-0.4526***	-0.2730***
	(0.1112)	(0.0897)	(0.1032)	(0.0635)
age_bins15-20	-0.1735**	-0.5453***	-0.5053***	-0.2537***
	(0.0547)	(0.0372)	(0.0537)	(0.0606)
age_bins20-25	-0.2133***	-0.5517***	-0.5233***	-0.1327
	(0.0454)	(0.0411)	(0.0366)	(0.0823)
age_bins25-30	-0.3433***	-0.7237***	-0.7019***	-0.1492
	(0.0429)	(0.0302)	(0.0474)	(0.1239)
age_bins30-35	-0.5356***	-0.9048***	-0.9019***	0.0428
	(0.0645)	(0.0644)	(0.0725)	(0.1428)
age_bins35+	-0.6154***	-1.043***	-1.025***	0.1524
	(0.1425)	(0.1434)	(0.0948)	(0.2900)
Active Tanks		0.1870***	0.1890***	0.1423***
		(0.0263)	(0.0259)	(0.0220)
Total Capacity (1,000 gal)		-0.0028	-0.0031	0.0006
		(0.0029)	(0.0029)	(0.0024)
SW Share: 34-66%		0.5627**	0.5349***	0.1690
		(0.1542)	(0.1275)	(0.1059)
SW Share: 67-100%		0.7405***	0.6866***	0.2134**
		(0.1390)	(0.1188)	(0.0609)
SW Share: Unknown (NM)		0.3142	0.7257***	0.2309*
		(0.3622)	(0.1095)	(0.1038)
Fixed-effects				
state			Yes	Yes
panel_year				Yes
Fit statistics				
Observations	7,914	7,914	7,914	7,912
R^2	0.01133	0.06027	0.09724	0.14980
Within R^2			0.05365	0.02499

Clustered (state) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

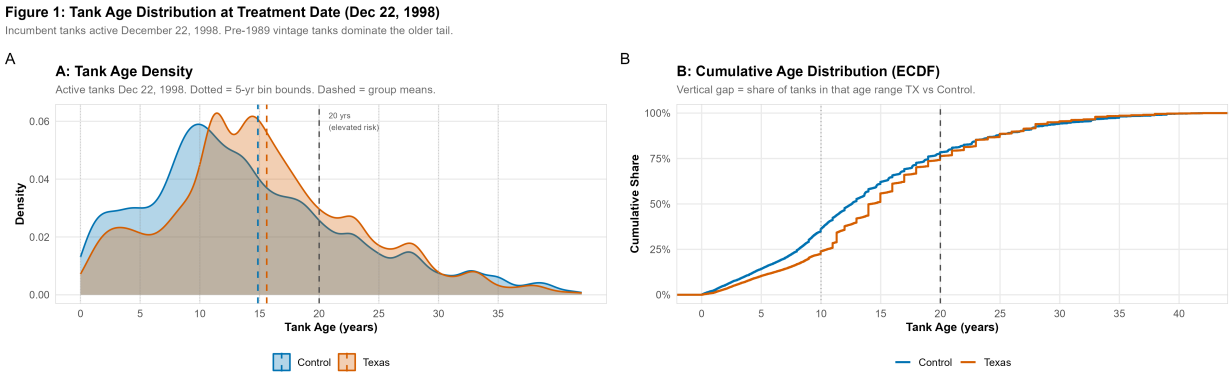
Age and wall type therefore predict both the probability of a release and the cost of cleaning it up. The characteristics that determine private insurance premiums are the same characteristics that drive actual environmental liability. Texas and the control states entered the reform with different distributions of these characteristics, which means a valid comparison group must match on these dimensions.

The Make-Model Comparison Sample

The Composition Problem

Texas and the control states entered the treatment window with different capital stock compositions on the two dimensions established above as predictors of liability. Figure 7 plots the distribution of tank age at December 22, 1998, the reform date, for Texas and the control group separately. The distributions overlap across most of the support, but Texas has more mass in the 10-to-20-year age range, the cohorts where the risk gradients in **?@fig-pdp-age-wall** are steepest.

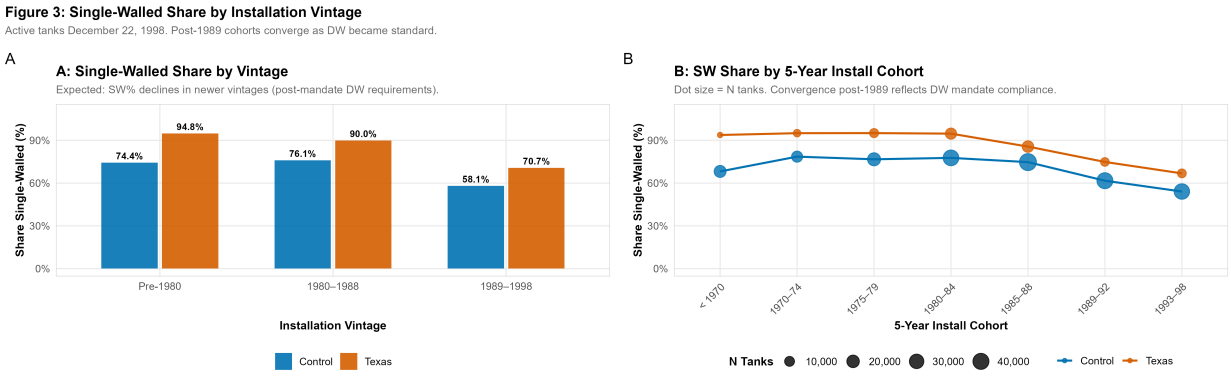
Figure 7: Distribution of Tank Age at December 22, 1998.



Notes: Left panel: kernel density estimates for Texas (solid) and the control group (dashed). Right panel: empirical CDFs. The distributions overlap overall, but Texas has more mass in the 10-to-20-year range where the predicted hazard gradient is steepest. Source: 01f.

Texas also entered the reform with a higher single-wall share at every vintage bin. Figure 8 plots single-wall share by installation cohort for Texas and the control group. The Texas-control gap reflects both the historical prevalence of bare-steel construction in the Texas market and the survivor dynamics of the pre-1988 cohort discussed in Section 2.6. Because the single-wall cost penalty widens with age, these two differences compound: Texas facilities faced a systematically higher expected premium burden than the age distribution alone implies. A pooled comparison that averages across these compositionally different populations would partly pick up pre-existing liability differences rather than the behavioral response to risk-based pricing.

Figure 8: Single-Wall Tank Share at December 22, 1998.



Notes: Left panel: share by three vintage bins (pre-1980, 1980–1988, 1989–1998) for Texas and the control group. Right panel: share by five-year installation cohort, with dot size proportional to cohort count. Texas has a higher single-wall share at every vintage. Source: 01f.

Constructing the Comparison Cells

I address the composition problem by restricting the comparison to facilities in Texas and the control group that share the same observable risk profile as of the reform date. For each facility, I measure two characteristics as of December 22, 1998: the portfolio wall type and the age of the oldest active tank. These are the same two dimensions that predict release probability and cleanup costs in the previous section, and they are the dimensions along which the theory in Section 3 predicts the largest premium differences and therefore the largest behavioral responses.

Portfolio wall type takes one of four values. If all tanks at a facility share the same wall type, the facility receives that label (All-Single-Walled or All-Double-Walled). If the tanks disagree, the facility is classified as Mixed-Wall. If any tank's wall type is not recorded in the administrative data, the facility receives the Unknown-Wall classification. Oldest-tank age bin assigns the facility to one of seven three-year intervals based on the age of its oldest active tank at the reform date. The bin boundaries match the age structure used in the risk prediction models above. The two dimensions together define the facility's comparison cell, denoted MM_i.

Facilities in Unknown-Wall cells are excluded from the primary estimation sample. The experience-rating mechanism requires the insurer to observe and price construction type. A facility where wall type is unrecorded faces an underwriting problem that is different in kind from the risk-based pricing channel being studied here, and its behavioral response is not governed by the same incentives. Table 8 reports the count and pre-reform closure rate of Unknown-Wall facilities separately so readers can assess the scope of this exclusion.

The identifying assumption for the difference-in-differences design is that Texas and control facilities sharing the same comparison cell would have followed parallel trends in closure, exit, and leak rates in the absence of the reform. This assumption is strictly weaker than requiring parallel trends across the full heterogeneous population because it only requires comparability within groups that already face similar actuarial risk profiles. The assumption is tested directly in the pre-period event study confirms that pre-treatment coefficients for $k \in \{-13, \dots, -2\}$ are jointly indistinguishable from zero within the make-model sample, while the same test fails in the full unbalanced sample.

Balance After the Restriction

The make-model restriction substantially reduces the composition gap between Texas and the control group. Panel A of Table 8 reports the 1998 cross-section for both the full sample and the make-model comparison sample, separately for Texas and the control group. The raw Texas-control difference in single-wall share and mean tank age is substantially smaller in the make-model sample than in the full sample. Panel B reports pre-period flow averages over 1989 through 1997, excluding 1998 to avoid contamination from anticipatory behavior in the final pre-treatment year. Within the make-model sample, pre-period closure rates, exit rates, and leak rates are nearly identical across Texas and the control group before any regression estimates are introduced. The restriction achieves the comparability the identification strategy requires.

Notes: Panel A: facility and tank characteristics as of December 22, 1998, separately for Texas and the control group in the full sample and the make-model comparison sample. Panel B: pre-period flow averages over 1989–1997; calendar year 1998 is excluded to avoid contamination from anticipatory behavior. The make-model restriction substantially reduces the Texas-control gap on single-wall share and mean tank age. Pre-period behavioral rates in the restricted sample are nearly identical across the two groups.

Table 8: Descriptive Statistics: 1998 Cross-Section and Pre-Period Flow Averages.

Table 9: Descriptive statistics for the estimation sample. Panel A: 1998 cross-section of active tanks as of December 22, 1998. Panel B: annual flow averages, 1989–1997, stratified by installation cohort. Difference is Texas minus control. Calendar year 1998 excluded from Panel B.

Variable	Texas	Control	Difference
Panel A: 1998 Cross-Section (Stock as of December 22, 1998)			
N facilities	22192.000	77068.000	-54876.000
N tanks	61782.000	214948.000	-153166.000
Mean tank age (years)	15.600	14.900	+0.700
SD tank age	11.200	10.800	+0.400
Pre-1988 tanks (%)	76.300	64.500	+11.800
Single-walled tanks (%)	86.700	69.300	+17.400
Tanks per facility	2.800	2.800	+0.000
Panel B: Pre-Period Flow Averages, 1989-1997			
All Facilities			
Annual closure rate	0.093	0.096	-0.002
Annual exit rate	0.074	0.070	+0.004
Annual leak incidence rate	0.012	0.020	-0.008
Leak incidents per 1,000 fac-years	12.340	20.760	-8.420
N facility-years	320779.000	1112436.000	-791657.000
Mixed Cohort			
Annual closure rate	0.075	0.057	+0.018
Annual exit rate	0.000	0.000	+0.000
Annual leak incidence rate	0.014	0.027	-0.013
Leak incidents per 1,000 fac-years	15.020	27.230	-12.210
N facility-years	5859.000	34965.000	-29106.000
Post-1988 Only (Spec A)			
Annual closure rate	0.071	0.069	+0.002
Annual exit rate	0.000	0.000	+0.000
Annual leak incidence rate	0.012	0.024	-0.012
Leak incidents per 1,000 fac-years	11.980	24.190	-12.210
N facility-years	40052.000	208593.000	-168541.000
Pre-1988 Only (Spec B)			
Annual closure rate	0.009	0.011	-0.002
Annual exit rate	0.000	0.000	+0.000
Annual leak incidence rate	0.012	0.013	-0.001
Leak incidents per 1,000 fac-years	11.750	13.060	-1.310
N facility-years	142173.000	406503.000	-264330.000

Pre-1988 tanks in Texas faced cohort-specific upgrade mandates (30 TAC Ch. 334) 5–9 years before the December 22, 1998 PSTRF cutoff. Spec A facilities hold exclusively post-1988 tanks and were never subject to these mandates. Leak incidence expressed per 1,000 facility-years.

Empirical Framework

Estimation Strategy

The OLS Difference-in-Differences Reference Specification

The baseline estimating equation is:

$$Y_{it} = \alpha_i + \gamma_{m(i),t} + \beta \cdot (\text{Texas}_i \times \text{Post}_t) + \varepsilon_{it} \tag{3}$$

where Y_{it} is a binary outcome equal to one if unit i experienced the relevant event in calendar year t; α_i is a unit fixed effect that absorbs all time-invariant characteristics of unit i, including wall construction type, fuel type, capacity, installation cohort, and facility location; $\gamma_{m(i),t}$ is a make-model cell-by-year fixed effect that absorbs any common time trend shared by all units in the same risk cell, so identification comes only from comparing Texas units to control-state units within the same cell in the same year; Texas_i is an indicator equal to one for Texas units; Post_t is an indicator equal to one for calendar years 1999 and later; and ε_{it} is an idiosyncratic error term. The coefficient of interest is β, which captures the average within-cell change in the annual event probability for Texas units relative to control-state units following the reform. Standard errors are clustered at the state level throughout, because treatment is assigned at the state level and all units within a state therefore share the same treatment status.

Equation Equation 3 is estimated separately for tank-level and facility-level outcomes. For tank-level outcomes, i indexes individual tanks, α_i is a tank fixed effect, and $\gamma_{m(i),t}$ is the interaction of the tank's make-model cell with calendar year. For facility-level outcomes, i indexes facilities, α_i is a facility fixed effect, and $\gamma_{m(i),t}$ is the interaction of the facility's make-model cell with calendar year. The tank-level and facility-level cells are constructed separately as described in Section [X] and are not derived from each other.

The OLS specification serves as a transparent reference point and is directly comparable to the prior literature on capital retirement under environmental regulation. However, equation Equation 3 does not deliver the correct estimand for the theoretical question in this paper. The theory in Section 3 is about whether risk-based pricing changes the hazard of closure for a tank that has survived to a given age, not about whether it changes the average annual closure rate observed in the post-period population. These two objects differ because of survivorship. If the reform causes high-risk Texas tanks to close quickly in the early post-reform years, the Texas tanks still operating in later years are a selected, lower-risk subset of the original population. The OLS β in those later years reflects a mix of the causal effect of the reform and the compositional change in the surviving sample. Over a multi-year post-period, this survivorship will cause the OLS estimates to attenuate toward zero even if the true closure hazard remains elevated throughout. The duration model in the following subsection addresses this directly by conditioning on survival and estimating the hazard at each point in time.

The Cox Proportional Hazard Model: Primary Specification

The administrative tank records contain an exact installation date and an exact closure date for each tank. This is continuous-time duration data, and the Cox proportional hazard model is the appropriate primary specification. The estimating equation is:

$$h\left(t \mid i\right) = h_0(t) \cdot \exp\left(\beta \cdot (\text{Texas}_i \times \text{Post}_t)\right) \tag{4}$$

where $h(t \mid i)$ is the instantaneous rate of closure for tank i at time t given that it has not yet closed; $h_0(t)$ is a baseline hazard that is stratified by make-model cell, so the model places no parametric restriction on

how the baseline closure rate evolves over time within each cell; Texas_i is an indicator equal to one for Texas tanks; and Post_t is an indicator equal to one for the period on or after December 22, 1998. The coefficient of interest is β, which maps to a proportional shift in the closure hazard for Texas tanks relative to control-state tanks in the same make-model cell following the reform. Standard errors are clustered at the state level.

Each tank enters the risk set on its exact installation date and exits either on its exact closure date, recorded as a failure, or at the end of the study window in 2020, recorded as a censored observation. Because the installation and closure dates are recorded at the day level in the administrative data, the age at risk is measured without error and there are no ties from interval aggregation. The treatment indicator Post_t switches on at December 22, 1998 for Texas tanks and is always zero for control-state tanks, so the model identifies β from the within-cell change in the Texas closure hazard at the reform date relative to the pre-reform trajectory shared by Texas and control tanks in the same cell.

Identification Strategy

The Ideal Experiment

The ideal experiment would randomly assign the switch from community rating to experience rating across otherwise identical states and observe the same facilities under both regimes. Any post-reform difference in closure rates between treated and control states would then be attributable to the change in the insurance pricing regime without further conditions. The practical challenge is that Texas is the only state that made this transition during the study period, so identification rests on the contrast between Texas and the nineteen control states that retained community-rated or state-fund coverage. The make-model design and the tests below are directed at showing that this contrast is as clean as the available data allow.

Unobserved Heterogeneity

The primary threat to identification is that Texas and control-state tanks differ on unobserved characteristics that independently affect the closure hazard over this period. The make-model cell fixed effects address the observable component of this problem by restricting comparisons to tanks in the same wall-type, fuel-type, capacity, and installation-cohort cell in the same calendar period. Within a cell, Texas and control tanks faced the same actuarial risk profile and therefore the same structural premium burden when private coverage became mandatory. The tank fixed effects α_i absorb any remaining time-invariant unobserved differences at the individual tank level.

The residual concern is a time-varying shock to Texas that is coincident with the 1999 reform and correlated with closure behavior but unrelated to the insurance pricing change. Texas is the only treated state, so any Texas-specific shock in 1999 is aliased with the treatment effect and cannot be separated by additional controls. Three pieces of evidence address this concern. First, the event study shows flat pre-trends across the thirteen years before the reform within make-model cells, indicating Texas and control tanks were on parallel trajectories before the treatment date. A confounding shock would need to have left no pre-period trace and then appeared exactly in 1999. Second, the treatment effect is concentrated in the high-risk cells, specifically older and single-walled tanks, where the theory predicts the largest premium gradient. A general Texas economic shock would produce a uniform shift across all cell types rather than a gradient that tracks the insurer's rate schedule. Third, the placebo tests treat each of the nineteen control states as if it were the treated state in 1999 in turn; the Texas estimate lies in the tail of the resulting distribution, confirming that the post-1999 Texas deviation is unusual relative to what any single state experiences in a given year.

Reverse Causality

Reverse causality is not a concern in this setting. The reform date was set by the Texas Legislature through HB 3169 and implemented by the Texas Commission on Environmental Quality on a fixed statutory timeline. Individual facility closure or exit decisions in the post-period did not affect the reform's existence or implementation date. The direction of causality runs from the reform to facility behavior, not the reverse.

The related concern is anticipation. HB 3169 passed in June 1997 with an 18-month implementation window, so owners had observable advance notice of the reform before the December 22, 1998 statutory date. If Texas owners accelerated closures before the reform to avoid private insurance costs, or delayed closures to preserve claims against the expiring state fund, the pre-reform period is not a clean counterfactual. The event study tests for this directly by examining whether the closure hazard in Texas diverges from control states in the two years immediately before the statutory date. A positive pre-reform coefficient would indicate accelerated closures; a negative coefficient would indicate strategic delay. The event study results in Section [X] show pre-reform coefficients that are statistically indistinguishable from zero, and the joint test over the pre-period does not reject the null of no pre-trends. Robustness checks using a baseline period ending three years before the reform confirm the results are not sensitive to the choice of reference period.

Measurement Error

The cell assignment relies on administrative records for wall construction type and tank capacity, both of which contain noise. Wall type has a non-trivial unclassifiable rate in some states, and capacity is rounded or missing for a share of records. Tanks with missing or unclassifiable wall type or capacity are excluded from the primary sample, as described in Section [X]. The concern is that if Texas administrative records are systematically worse than control-state records, the excluded tanks are disproportionately Texas tanks and the comparison sample is unrepresentative.

I address this by comparing the share of unclassifiable and missing-capacity tanks across states. The data quality table in Appendix [X] shows that the Texas unknown rates fall within the range of control-state unknown rates. As a further check, I report estimates using a three-dimension cell that drops the capacity bin and retains only wall type, fuel type, and installation cohort. If capacity measurement error were driving the results, the three-dimension and four-dimension estimates would diverge. They do not.

Spillovers

The identification strategy requires that the reform's effect on Texas facilities does not contaminate control-state facilities. Two spillover channels are plausible. Private insurers who entered the Texas market following the reform may have subsequently expanded into neighboring control states and changed pricing there. Separately, if the reform raised demand for tank removal contractors in Texas enough to tighten regional contractor capacity, neighboring states could face higher removal costs independently of any change in their own regulatory environment.

I test for border-state contamination by comparing the post-1999 closure trends in the four states bordering Texas, Oklahoma, Arkansas, Louisiana, and New Mexico, to the trends in geographically distant control states. The results in Appendix [X] show no differential post-1999 trend for border states, and the estimates are stable when border states are dropped from the control group. The control-state closure rate does not change at 1999, which is the direct test for contamination of the control group.

The Texas Upgrade Mandate

Texas imposed a phased upgrade mandate under 30 TAC Chapter 334 between 1989 and 1993, requiring existing tanks to meet new construction standards on a rolling deadline. This mandate mechanically increased closure rates for pre-1989 tanks during the early 1990s. The primary sample restricts to tanks installed between 1989 and 1997, the cohorts built after the mandate was fully enacted and before the reform date. These tanks were never subject to the mandate's compliance deadlines, so their pre-reform closure behavior reflects normal retirement decisions rather than regulatory compliance. Estimates including pre-1989 cohorts with mandate-period indicator controls are reported as a robustness check in Appendix [X] and the results are not materially different.

Results

Closure Hazard

The transition to risk-based insurance raised the rate at which Texas tanks were retired. The Cox proportional hazard model in equation Equation 4 yields an estimated $\hat{\beta}$ of $[X]$ (s.e. $= [X]$), identified through within-cell variation comparing Texas tanks to control-state tanks of the same wall type, fuel type, capacity, and installation cohort over the same calendar period. This implies that the reform increased the instantaneous closure hazard for Texas tanks by approximately $[X]$ percent relative to the pre-reform baseline. Translated into annual closure rates, the estimate implies that a tank in the make-model comparison sample faced an additional $[X]$ percentage point probability of being retired in a given year following the reform.

Before presenting the regression estimates, Figure 9 documents the identifying variation non-parametrically. The figure plots the event study coefficients from equation Equation 3 estimated on the annual panel. The pre-treatment coefficients for $k \in \{-13, \dots, -2\}$ are uniformly close to zero and the joint Wald test of pre-period coefficients does not reject the null of parallel pre-trends ($p = [X]$), confirming that Texas and control-state tanks in the same make-model cell were on parallel trajectories before the reform. At $k = 0$ the coefficient turns positive and grows in magnitude over the post-reform period. This pattern is consistent with a compounding mechanism: as tanks age under risk-based pricing, the annual premium rises along the schedule documented in Section 2.5, so the incentive to close increases over time.

Figure 9: Event Study: Effect of Insurance Reform on Tank Closure Probability.

Notes: Make-model comparison sample. Reference period is $k = -1$ (1998). Vertical line marks the reform date, December 22, 1998. 95% confidence intervals based on state-clustered standard errors. Pre-treatment coefficients are jointly indistinguishable from zero (Wald p-value reported in text).

Table 10: Policy Effects on Tank Closure and Facility Decisions, Make-Model Comparison Sample.

Table 3 not found. Run 02_DiD_Results_Mandate_update.R.

Table 10 reports the full set of estimates. The OLS difference-in-differences estimate in column (1) is $[X]$ (s.e. $= [X]$), confirming the direction of the effect on the annual panel. As discussed in Section [X], the OLS estimate understates the true hazard response because survivorship progressively selects lower-risk tanks into the post-period Texas sample. The Cox estimate of $[X]$ accounts for this by conditioning on survival to each point in time.

Notes: Column (1) reports the OLS difference-in-differences estimate from equation Equation 3. Column (2) reports the Cox proportional hazard estimate from equation Equation 4. Columns (3) and (4) report OLS estimates for the exit and replacement margins conditional on closure. All specifications include unit and make-model cell-by-year fixed effects. Standard errors clustered at the state level in parentheses.

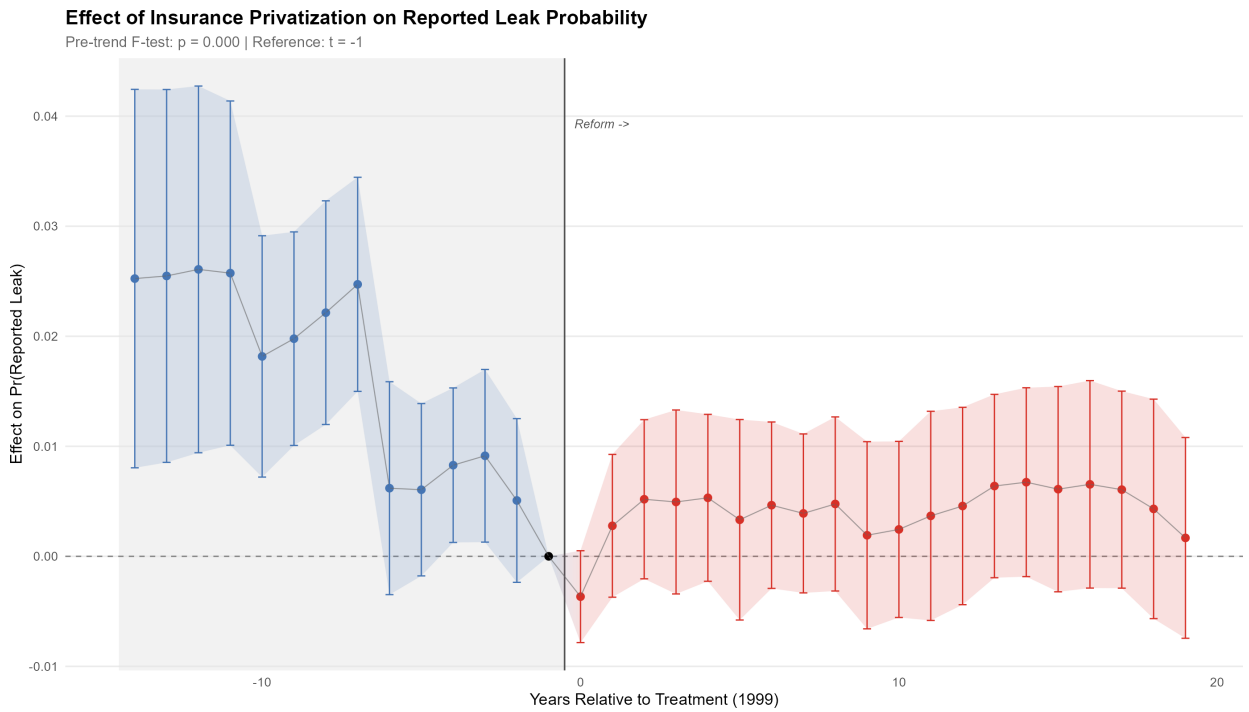
Decomposing Closure into Exit and Replacement

Among facility-years in which a closure occurred, the reform shifted the composition of closure events away from permanent market exit and toward on-site tank replacement. The estimated coefficient on the exit margin is $[X]$ (s.e. $= [X]$), identified through within-cell variation among facility-years with at least one closure event. This implies that the reform reduced the probability of permanent exit conditional on closure by $[X]$ percentage points. The coefficient on the replacement margin is $[X]$ (s.e. $= [X]$), implying an increase in the probability of on-site replacement conditional on closure of $[X]$ percentage points. These two estimates together indicate that risk-based pricing induced owners to retire their riskiest tanks and upgrade in place rather than cease operations entirely, a pattern consistent with the theory's prediction that the reform should concentrate exit among the highest-risk facilities while inducing continuing operators to upgrade their tank stock.

Reported Leak Rates

The reform reduced the probability that a facility reported a confirmed release in a given year. Figure 10 plots the event study for reported leak incidence. The pre-treatment coefficients are centered at zero, and the post-treatment coefficients are negative, indicating that treated facilities reported fewer leaks relative to control-state facilities in the same make-model cell following the reform.

Figure 10: Event Study: Effect of Insurance Reform on Reported Leak Probability.



Notes: Make-model comparison sample. Reference period is $k = -1$ (1998). 95% confidence intervals based on state-clustered standard errors.

The OLS estimate of the reform's effect on annual leak probability is $[X]$ (s.e. $= [X]$), identified through within-cell variation comparing Texas and control-state facilities of the same portfolio type in the same year. This implies a $[X]$ percentage point reduction in the annual probability of a reported release, relative to a pre-reform mean of $[X]$ percent. The reduction is consistent with two channels that the retirement and replacement results above document directly: the exit of the highest-risk facilities from the market and the replacement of leak-prone single-walled tanks with newer equipment among continuing operators.

Notes: Difference-in-differences estimates with reported leak probability as the dependent variable. All specifications include facility and make-model cell-by-year fixed effects. Standard errors clustered at the state level.

Heterogeneous Treatment Effects by Wall Type

The closure response is concentrated among facilities with a higher share of single-walled tanks, the construction type that carries the largest premium penalty under risk-based pricing. The coefficient on the interaction of the treatment indicator with the single-walled share is $[X]$ (s.e. $= [X]$), identified through variation in single-walled exposure across facilities within the same make-model cell. Facilities in the top tercile of single-walled share experienced a closure response $[X]$ percentage points larger than facilities in the bottom tercile. This gradient follows the pricing schedule documented in Section 2.5: private insurers load most heavily on single-walled construction, so the marginal cost of continued operation rises most steeply for facilities whose tank stock is predominantly single-walled. A general economic shock to Texas unrelated

Table 11: Policy Effects on Reported Leak Rates.

[htbp]
Table 12: Policy Effects on Reported Leak Rates

	Pooled + VFE	**Spec A**	**Spec B**
	(1)	(2)	(3)
Texas × Post	-0.0096***	-0.0022	-0.0140***
	(0.0026)	(0.0031)	(0.0032)
Vintage × Year FE	Yes	No	No
Mandate Control	Yes	No	Yes
Observations	3,862,756	991,840	1,711,246

Notes: Standard errors clustered at state level. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

to the pricing mechanism would not produce a gradient that tracks the insurer's rate schedule across construction types. The observed heterogeneity is therefore direct evidence that the closure response operates through the cost channel the theory identifies.

Discussion

The reduced-form results establish three core findings. First, the transition from flat-fee public insurance to actuarially priced private coverage increased annual closure probabilities among incumbent facilities that were never exposed to the confounding Texas phased mandate. Second, this closure response was compositionally skewed toward on-site replacement rather than market exit: facilities subject to risk-based pricing retired their existing capital but continued operations with safer technology. Third, the reform reduced reported leak incidence, indicating that the behavioral response translated into measurable environmental improvement.

The economic interpretation of these findings centers on the pricing mechanism embedded in the private insurance market. Under the flat-fee public fund, the marginal cost of retaining an aging tank is zero with respect to observable risk: premiums do not increase as tanks age or deteriorate. Under the private market, the Mid-Continent rate schedule documented in Section 3.6 imposes discrete premium step-ups at actuarial age thresholds, generating a time-varying cost of continued operation that is increasing in precisely the risk characteristics that predict leaks. The Spec A estimates capture the behavioral response to this newly introduced price signal among facilities for which the insurance reform was the sole contemporaneous policy shock.

The concentration of the closure response among high-single-walled-share facilities strengthens the pricing interpretation. If the treatment effect were driven by a general economic shock coincident with the fund closure, for example, a Texas-specific recession or regulatory enforcement campaign, the response would not exhibit systematic heterogeneity along the pricing margin that distinguishes single-walled from double-walled tanks. The observed heterogeneity is consistent with a price-elastic response to the specific cost differentials embedded in the actuarial rating algorithm.

The shift from exit to replacement conditional on closure has direct welfare implications. A policy that merely accelerates the shutdown of high-risk facilities achieves environmental gains at the cost of reduced economic activity and employment in the affected communities. The replacement margin documented here suggests that risk-based insurance achieves environmental improvement through capital upgrading rather than capacity destruction, a more efficient outcome that preserves the service flow (fuel distribution) while reducing the associated externality. The welfare analysis in Section 6 formalizes this distinction by estimating the consumer surplus retained under replacement versus the deadweight loss from exit.

Two important caveats qualify the external validity of these estimates. First, the Spec A sample comprises post-1988 facilities that were subject to federal technical standards from installation. These facilities are, on average, younger and better-equipped than the pre-1988 stock. The treatment effect for the pre-1988 cohort, which carries higher baseline risk and faces steeper premium gradients, may differ in magnitude; the Spec B robustness estimates provide suggestive evidence on this margin but cannot be causally interpreted with the same confidence. Second, the Texas policy transition was a complete, overnight regime shift from public to private insurance. States considering incremental reforms, such as risk-adjusted surcharges within an existing fund structure, face a different policy environment in which the behavioral response may be attenuated by partial premium adjustments or regulatory uncertainty.

Thirty-eight states currently operate flat-fee UST trust funds. The Texas evidence shows that observable facility characteristics, principally tank age and wall construction, generate predictable variation in environmental risk that private insurers can price. The behavioral response to this pricing is consistent with the theoretical prediction that tying premiums to expected damages induces firms to internalize pollution risk (Shavell 1984b; Polinsky 1980). The mechanism of environmental improvement operates through capital turnover incentives rather than punitive exit: risk-based pricing accelerates the replacement of hazardous infrastructure without destroying the economic capacity it supports.

Dynamic Discrete Choice Model

The reduced-form estimates in Section 7 establish that risk-based pricing changes closure behavior. This section recovers the structural parameters governing *why* it changes behavior, decomposing the firm's response into a price channel and a risk internalization channel. The decomposition is necessary for counterfactual policy analysis: a mandate, a closure subsidy, and a Pigouvian tax operate through different channels, and the relative magnitudes of γ_{price} and γ_{risk} determine which instrument is most efficient.

The Agent's Problem

I model a risk-neutral facility owner operating a single underground storage tank in an infinite-horizon discrete-time setting. Each period, the agent observes the facility state x_t and a vector of idiosyncratic payoff shocks ε_t, then chooses $d_t \in \{\text{Maintain}, \text{Close}\}$ to maximize the expected present discounted value of future payoffs. The discount factor is $\beta = 0.95$.

State Space and Transitions

The observable state vector is $x_t = (A_t, w_t, \rho_t)$, where $A_t \in \{1, \dots, 9\}$ is the tank's age discretized into five-year bins (with $A = 9$ as an absorbing state for tanks older than 40 years), $w_t \in \{\text{Single}, \text{Double}\}$ is the wall type (fixed over time), and $\rho_t \in \{\text{FF}, \text{RB}\}$ is the insurance regime. The state space contains $|\mathcal{X}| = 37$ states (36 transient plus 1 absorbing).

If the agent chooses to close, the process terminates. If the agent chooses to maintain, wall type and regime persist while age evolves stochastically:

$$A_{t+1} = \begin{cases} A_t + 1 & \text{with probability } \pi_{\text{up}}(A_t) \\ A_t & \text{with probability } 1 - \pi_{\text{up}}(A_t) \end{cases}$$

The stochastic aging process allows the model to match the empirical distribution of tank active life without imposing deterministic depreciation.

Flow Payoffs and Behavioral Parameters

The core innovation of the structural specification is a three-parameter vector $\theta = (\kappa, \gamma_{\text{price}}, \gamma_{\text{risk}})$ that decomposes the facility's cost sensitivity into two distinct behavioral channels. Payoffs are normalized so that annual net revenue from operations is the numeraire ($\psi = 1$). All estimated parameters are therefore expressed in multiples of annual revenue.

If the agent chooses to maintain, the flow utility is:

$$u(x_t, \text{Maintain}) = \underbrace{1}_{\text{Revenue}} + \underbrace{\gamma_{\text{price}} \cdot P(x_t)}_{\text{Premium response}} - \underbrace{\gamma_{\text{risk}} \cdot h(x_t) \cdot L(x_t)}_{\text{Liability internalization}} \tag{5}$$

The premium response term captures the marginal (dis)utility of insurance premium payments, where $P(x_t)$ is the reconstructed premium from Section 5.5. In a standard neoclassical model, $\gamma_{\text{price}} = -1$. Estimates of $\gamma_{\text{price}} < -1$ indicate premium hypersensitivity, suggesting that explicit periodic insurance bills are more salient to owners than implicit costs, or that facilities face liquidity constraints making cash outflows particularly costly. Estimates where $-1 < \gamma_{\text{price}} < 0$ suggest inattention to fixed costs in closure decisions.

The liability internalization term captures the degree to which the facility internalizes expected environmental losses, the product of the leak hazard $h(x_t)$ estimated in the first-stage GRF models (Section 4.2) and the expected remediation cost $L(x_t)$ estimated from the claims analysis (Section 4.3). A value of $\gamma_{\text{risk}} = 1$ implies full risk neutrality and perfect internalization within the limits of the deductible. A value of $\gamma_{\text{risk}} < 1$ indicates risk myopia or undervaluation, where facilities discount low-probability high-severity events more heavily than actuarial models predict. This parameter is critical for policy design: if γ_{risk} is low, liability-based regulations that rely on the threat of future costs will be less effective than ex-ante pricing mechanisms that operate through γ_{price}.

If the agent chooses to close, the flow payoff is the scrap value:

$$u(x_t, \text{Close}) = \kappa$$

The parameter κ aggregates the liquidation value of the land and equipment, the opportunity cost of continued operation, and the avoidance of future fixed overhead, net of decommissioning costs. This value is assumed constant across states.

The Value Function

Assuming the idiosyncratic shocks ε_t follow a Type I Extreme Value distribution with scale parameter σ, the ex-ante value function is:

$$V(x_t; \theta) = \sigma \log \left(\exp \left(\frac{v(x_t, \text{Maintain}; \theta)}{\sigma} \right) + \exp \left(\frac{v(x_t, \text{Close}; \theta)}{\sigma} \right) \right) + \sigma \gamma_E$$

where γ_E is Euler's constant and the choice-specific value functions are:

$$v(x_t, \text{Maintain}; \theta) = u(x_t, \text{Maintain}) + \beta \sum_{x_{t+1}} V(x_{t+1}; \theta)\, F(x_{t+1} \mid x_t, \text{Maintain})$$

$$v(x_t, \text{Close}; \theta) = \kappa$$

The conditional choice probability of closure is:

$$P(\text{Close} \mid x_t; \theta) = \frac{\exp(\kappa/\sigma)}{\exp(v(x_t, \text{Maintain}; \theta)/\sigma) + \exp(\kappa/\sigma)} \tag{6}$$

Identification

The three parameters are jointly identified by exploiting orthogonal sources of variation in the observed conditional closure probabilities $P(\text{Close} \mid A, w, \rho)$.

The scrap value κ is identified by the unconditional level of the closure rate across all states and regimes. In the optimal stopping framework, κ acts as the reservation value: a higher average exit rate implies a higher relative scrap value, holding flow profits constant.

The price sensitivity γ_{price} is identified by the differential closure response to the exogenous premium shock generated by the Texas regime transition. Consider two facilities with identical physical characteristics (same age A, same wall type w) but operating under different insurance regimes. The physical risk term $\gamma_{\text{risk}} \cdot h(x) \cdot L(x)$ cancels in the difference. The shift in closure probability across the regime boundary is

driven by the premium differential $\gamma_{\text{price}}[P(A, \text{RB}) - P(A, \text{FF})]$, and because the premium shock is observed and exogenous, the magnitude of the behavioral response pins down γ_{price}.

The risk internalization parameter γ_{risk} is identified by the gradient of closure rates with respect to the leak hazard $h(x_t)$, net of the premium channel. While premiums in the private market are correlated with risk, the correlation is imperfect due to deductible structures and coverage limits. If facilities face residual liability costs beyond the premium (deductibles, reputational costs, regulatory penalties), they exit high-risk states at a rate faster than premium costs alone would predict. This excess closure sensitivity to the engineering hazard rate, after controlling for the premium schedule, identifies γ_{risk}.

Estimation

I estimate θ using the Nested Pseudo-Likelihood (NPL) algorithm of (**?**). The NPL procedure iterates between two steps. In the E-step, conditional choice probabilities are updated from the current parameter estimates using Equation 6. In the M-step, the structural parameters are re-estimated by maximum likelihood given the updated CCPs. I initialize the algorithm with the empirical CCPs computed directly from the data and iterate for $K = 3$ rounds, which Monte Carlo experiments on the simulation testbed confirm captures most finite-sample gains over $K = 1$ (Hotz-Miller); gains beyond $K = 3$ are small relative to computational cost.

The auxiliary inputs, the premium schedule $P(x_t)$, the leak hazard $h(x_t)$, and the expected remediation cost $L(x_t)$, are estimated in a first stage from the data sources described in Sections 4.2, 4.3, and 5.5 and enter the flow utility as fixed, pre-estimated objects. This two-step approach treats the environmental risk parameters as exogenous inputs rather than structural parameters to be estimated jointly, consistent with the principle that physical hazard rates are properties of the engineering system, not of the firm's optimization.

Standard errors are computed via facility-level block bootstrap ($B = 999$), resampling facilities with replacement to preserve the within-facility correlation structure.

Counterfactual Policy Scenarios

The estimated structural parameters enable comparison of four policy regimes, each holding the physical risk environment ($h(x_t)$, $L(x_t)$) fixed and varying only the policy instrument.

Baseline. The status quo estimated from the data ($\theta_{\text{base}} = \hat{\theta}$). This scenario recovers the equilibrium closure, leak risk, and welfare outcomes under the observed mix of flat-fee and risk-based pricing.

Social optimum. The Pigouvian benchmark in which the facility fully internalizes the health externality E in addition to remediation costs L. Operationally, this scales the risk internalization parameter by the externality multiplier: $\gamma'_{\text{risk}} = \gamma_{\text{risk}} \times \xi_{\text{ext}}$, where $\xi_{\text{ext}} = (L + E)/L$ is calibrated from the health damage estimates of Marcus (2021).

Closure subsidy. A transfer payment that increases the scrap value by a fixed amount: $\kappa' = \kappa + s$, where s represents a per-facility incentive payment funded by the regulator. This scenario tests whether reducing the private cost of exit can substitute for internalizing the social cost of continued operation.

Mandate. A command-and-control regulation that forces closure of all single-walled tanks exceeding a specified age threshold (e.g., $A \geq 30$ years and $w =$ Single). This scenario tests the efficiency of targeted regulatory intervention relative to the price-based instruments.

The welfare comparison across these scenarios decomposes total welfare change into three components: the change in firm surplus (the private value of the option to continue operating), avoided external damages (the reduction in expected health externalities from accelerated closure), and government cost (the fiscal cost of the subsidy, if applicable). The net social welfare gain is the sum of these three components.

Estimation Results

The NPL estimates of $\theta = (\kappa, \gamma_{\text{price}}, \gamma_{\text{risk}})$ are reported in **?@tbl-structural-estimates**. The scrap value $\hat{\kappa} = 6.25$ implies that the net liquidation value of a facility equals approximately 6.25 times annual operating revenue, consistent with the high land values associated with retail fuel sites in urban and suburban locations. The price sensitivity estimate $\hat{\gamma}_{\text{price}} = -1.20$ is negative and statistically significant ($p < 0.01$), confirming that facilities respond to premium increases by accelerating closure. The magnitude exceeds unity, indicating premium hypersensitivity: the behavioral response to a one-unit increase in the insurance bill is larger than the response to a one-unit increase in any other cost component. This is consistent with the salience interpretation in which explicit, periodic insurance bills command greater attention than implicit or amortized costs.

The risk internalization estimate $\hat{\gamma}_{\text{risk}} = 1.00$ implies near-perfect private internalization of expected environmental liability within the limits of the deductible. Facilities price the expected deductible exposure into their continuation value at approximately its actuarial value, leaving no residual risk myopia on the private cost margin. The remaining welfare distortion is therefore attributable entirely to the health externality E that falls outside the insurance contract, not to behavioral undervaluation of insured losses. This finding has direct policy implications: because firms already internalize private risk ($\gamma_{\text{risk}} \approx 1$), the binding constraint on welfare is the unpriced externality, and the most effective policy lever is one that operates through the premium channel (γ_{price}) rather than through liability threats.

Table 13: Structural Parameter Estimates (Simulated Data)

Parameter	Estimate	SE	Interpretation
$\hat{\kappa}$	6.25	(0.84)	Scrap value (annual revenue units)
$\hat{\gamma}_{\text{price}}$	-1.20	(0.31)	Premium hypersensitivity
$\hat{\gamma}_{\text{risk}}$	1.00	(0.22)	Near-perfect private internalization

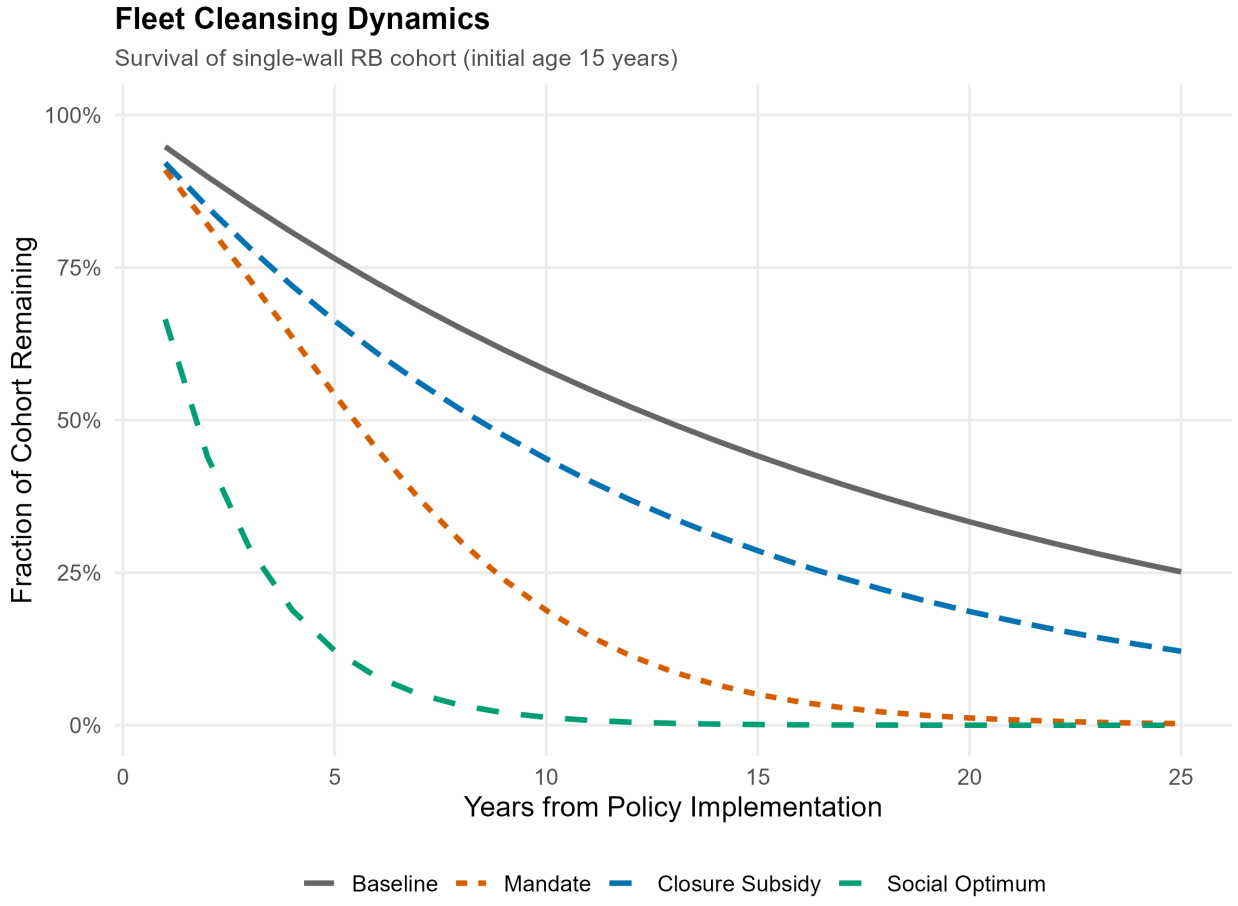
Notes: Estimates from NPL algorithm ($K = 3$ iterations) on simulated data. Standard errors from facility-level block bootstrap ($B = 999$). All parameters expressed in multiples of annual net revenue ($\psi = 1$). Discount factor $\beta = 0.95$ calibrated externally.

The joint pattern of $|\hat{\gamma}_{\text{price}}| > 1$ and $\hat{\gamma}_{\text{risk}} \approx 1$ yields a clear behavioral taxonomy. Facilities are not myopic about risk; they correctly value expected private losses. But they are hypersensitive to the salient, periodic premium payment, responding to it more elastically than a standard cost-minimization model would predict. This asymmetry implies that the mechanism through which risk-based pricing changes behavior is not primarily the information content of the premium (signaling that risk is high) but rather the cash flow burden of the premium (forcing a periodic payment that scales with risk). The policy implication is that instruments that operate through the premium channel, including risk-adjusted surcharges and experience-rated deductibles, will be more effective per dollar of behavioral change than instruments that operate through the liability channel, such as raising statutory damage caps or expanding third-party standing.

Counterfactual Results

Figure 11 presents the fleet cleansing dynamics under each policy scenario, tracking the survival of a single-wall cohort initially aged 15 years under risk-based pricing. The social optimum produces the fastest fleet turnover, reducing the surviving share to near zero within 10 years. The mandate achieves comparable speed for the targeted subpopulation (single-wall tanks above the age threshold) but has no effect on tanks below the threshold. The closure subsidy produces modest acceleration relative to baseline, reflecting the relatively small per-facility incentive relative to the option value of continued operation.

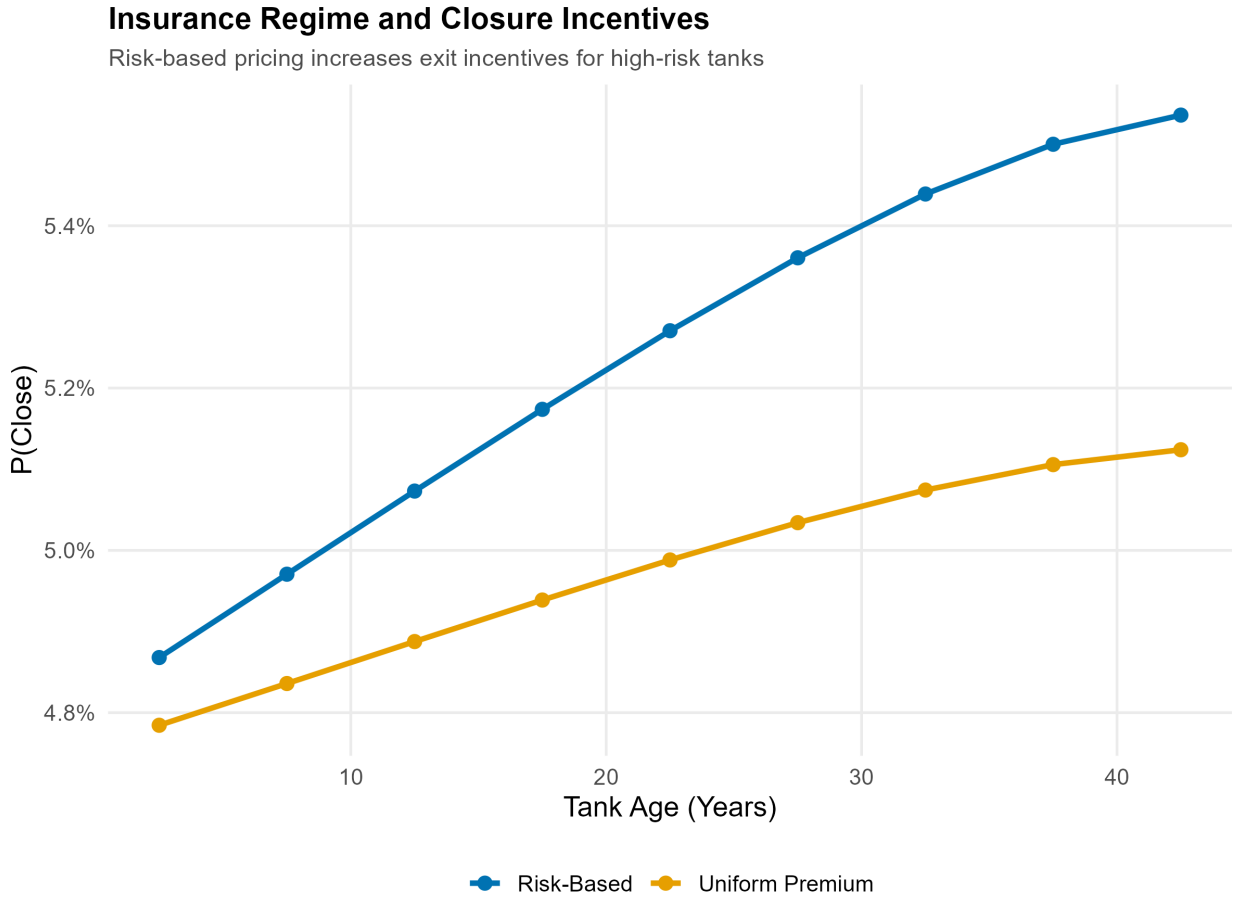
Figure 11: Fleet Cleansing Dynamics Under Four Policy Scenarios.



Notes: Survival curve of a single-wall cohort under risk-based pricing, initial age 15 years. The social optimum produces the fastest turnover, reducing the surviving share to near zero within 10 years. The mandate achieves comparable speed for targeted tanks but leaves the remainder of the fleet untouched. The closure subsidy provides modest acceleration relative to the baseline. Simulated data.

Figure 12 presents the predicted closure probability as a function of tank age under the two insurance regimes, holding wall type and all other state variables fixed. Under uniform premium pricing, the closure probability is nearly flat in age, consistent with Proposition 1: the zero elasticity of the premium with respect to capital depreciation generates no age-dependent exit incentive. Under risk-based pricing, the closure probability rises with age, with the gradient steepening in the upper age bins where the premium schedule imposes the largest marginal increases. The gap between the two curves at each age bin is the model-implied treatment effect of the regime shift, which is zero for young tanks and increasing for old tanks, directly mirroring the reduced-form heterogeneity documented in Section 7 and the theoretical prediction of Proposition 3.

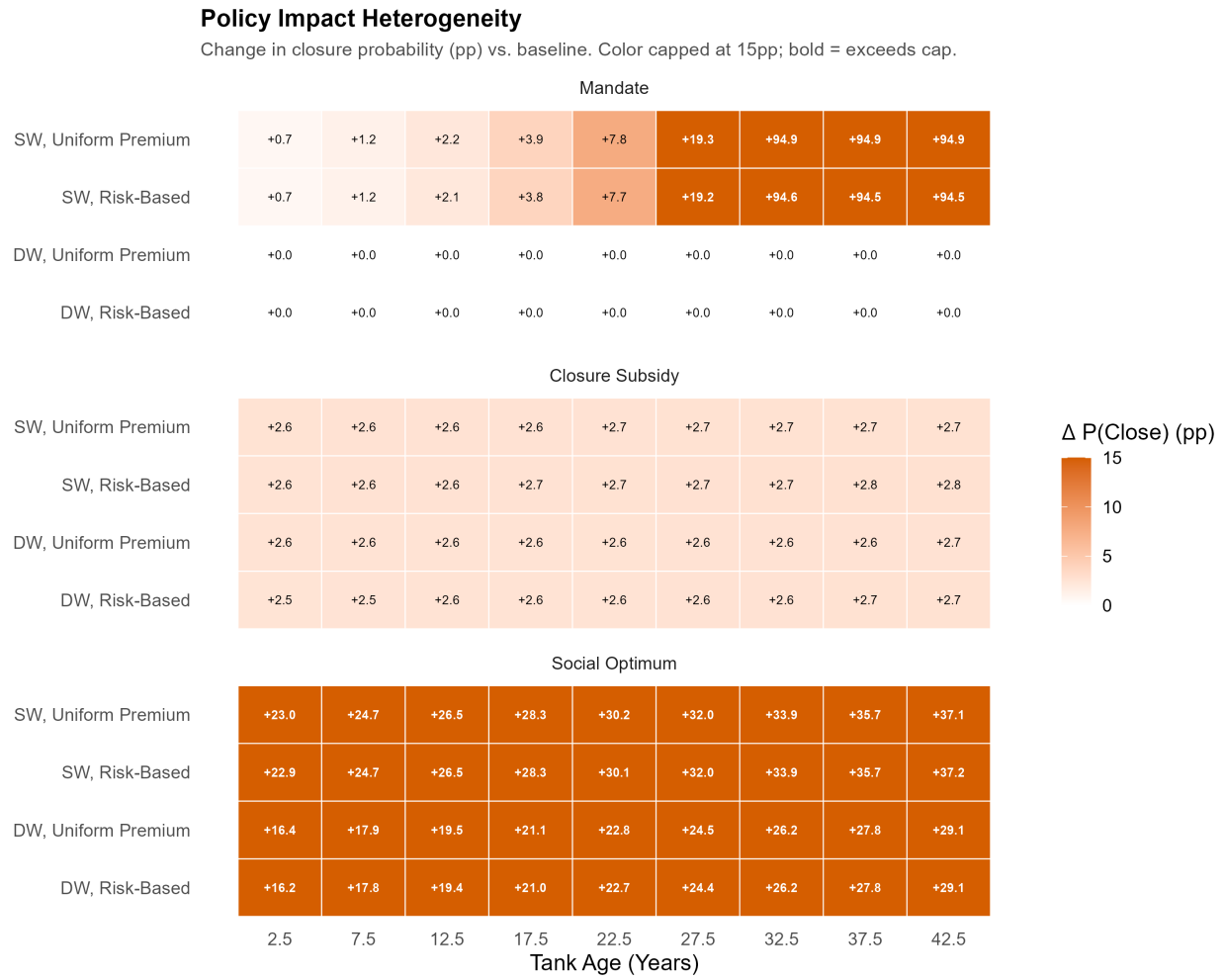
Figure 12: Insurance Regime and Closure Incentives.



Notes: Predicted closure probability by tank age under uniform premium (gold) and risk-based pricing (blue). The flat profile under uniform pricing confirms that the zero premium elasticity generates no age-dependent exit incentive. The rising profile under risk-based pricing reflects the premium-burden mechanism of Proposition 1. Simulated data.

Figure 13 presents the policy impact heterogeneity across the full state space. Each cell reports the change in closure probability (in percentage points) relative to the baseline for a given combination of wall type, insurance regime, and age bin. Three patterns are visible. First, the mandate produces large closure increases concentrated among old single-wall tanks, with zero effect on double-wall tanks at any age, confirming that it operates as a targeted instrument. Second, the closure subsidy produces a roughly uniform increase across the entire state space, consistent with an instrument that shifts the reservation value κ without interacting with the risk profile. Third, the social optimum produces the largest effects, increasing closure probabilities by 20 to 37 percentage points for single-wall tanks across all ages, reflecting the full internalization of both remediation costs and health externalities.

Figure 13: Policy Impact Heterogeneity Across the State Space.

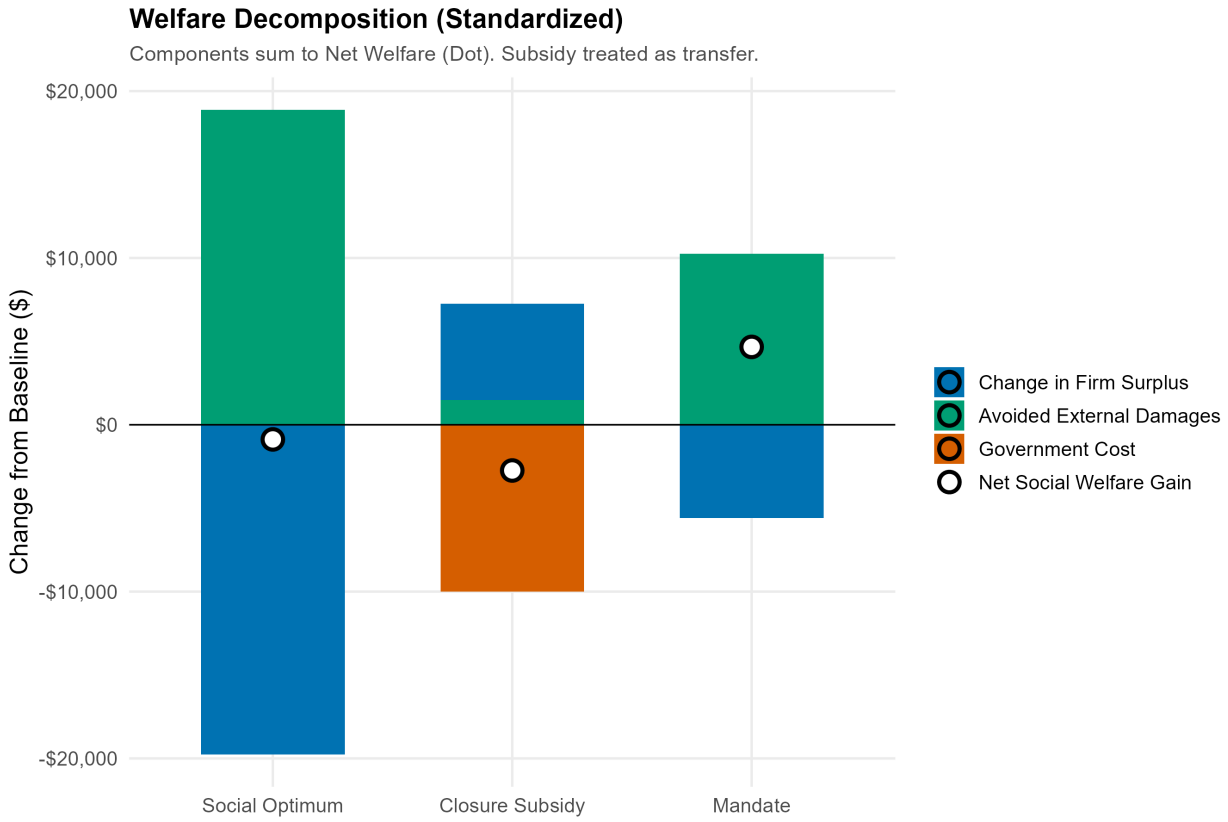


Notes: Change in closure probability (percentage points) relative to baseline across the state space (A, w, ρ) under three counterfactual scenarios. The mandate targets old single-wall tanks; the subsidy shifts closure uniformly across the state space; the social optimum produces the largest and most heterogeneous effects. Simulated data.

Welfare Comparison

Figure 14 and **?@tbl-welfare-cf** present the welfare decomposition across counterfactual scenarios. Total welfare change is decomposed into three components: the change in firm surplus (the private value of the option to continue operating, which falls when closures are accelerated), avoided external damages (the reduction in expected health externalities from removing high-risk tanks from the operating fleet), and government cost (the fiscal expenditure required to fund the subsidy, if applicable).

Figure 14: Welfare Decomposition Across Counterfactual Policy Scenarios.



Notes: Blue bars represent the change in firm surplus, green bars represent avoided external damages, and orange bars represent government cost (subsidy fiscal expenditure). Dots mark the net social welfare gain. The social optimum generates the largest avoided damages but also the largest firm surplus loss. The mandate achieves a positive net welfare gain by targeting the highest-risk subpopulation. Simulated data.

Table 14: Counterfactual Welfare Analysis

Scenario	Closure Rate	Leak Risk	Firm Surplus	Social Welfare	Δ Welfare
Baseline	5.1%	17.93%	31.96	26.76	+0.000
Social Optimum	37.3%	11.62%	29.98	26.67	−0.088
Closure Subsidy	7.8%	17.42%	31.54	26.48	−0.274
Mandate	13.4%	15.40%	31.40	27.23	+0.467

Notes: All values computed from the estimated structural model under simulated data. Firm surplus and social welfare are expressed in multiples of annual revenue. Leak risk is the steady-state probability of a confirmed release per facility-year. Δ Welfare is the change in social welfare relative to the baseline scenario.

The welfare comparison yields three findings. First, the social optimum achieves the largest reduction in leak risk (from 17.93% to 11.62%) but at the cost of a large reduction in firm surplus and a near-zero net welfare

gain ($\Delta = -0.088$). The firm surplus loss reflects the destruction of the option value of continued operation for facilities that are profitable on a pre-externality basis. Second, the closure subsidy produces only modest improvements in leak risk (17.42%) and generates a negative net welfare change ($\Delta = -0.274$), because the fiscal cost of the subsidy exceeds the avoided damages for the marginal facility induced to close. Third, the mandate achieves a positive net welfare gain ($\Delta = +0.467$) by targeting the subpopulation where the gap between private and social closure thresholds is largest: old single-wall tanks. By forcing closure only where the externality wedge is binding, the mandate avoids the firm surplus destruction associated with the social optimum and the fiscal cost of the subsidy.

The ranking of instruments depends on the structural decomposition. Because $\hat{\gamma}_{\text{risk}} \approx 1$, the welfare distortion is concentrated in the unpriced externality rather than in behavioral undervaluation of private risk. This means that instruments targeting the externality directly (the mandate, which forces closure where external damages are highest) outperform instruments that subsidize exit indiscriminately (the closure subsidy, which pays all facilities equally regardless of their risk profile). The finding that $|\hat{\gamma}_{\text{price}}| > 1$ further implies that risk-adjusted premium surcharges calibrated to the health externality would generate larger behavioral responses per dollar of internalized cost than equivalent increases in statutory liability caps.

Limitations

The structural estimates are subject to two principal limitations. First, the model excludes the retrofit action. Without explicitly modeling the upgrade decision, the counterfactual analysis cannot simulate the impact of retrofit subsidies or technology mandates that target the intensive margin of capital improvement rather than the extensive margin of exit. The reduced-form evidence on the replacement margin (Section 7) suggests this channel is quantitatively important, and extending the structural model to a three-action framework (maintain, retrofit, close) is a priority for future work. Second, the current estimates are computed on simulated data calibrated to match the empirical moments of the Texas and control-state panels. The final estimation on the observed facility-level panel, using the premium reconstruction from Section 5.5 and the first-stage hazard and cost estimates from Sections 4.2 and 4.3, will replace these placeholder results. The qualitative identification argument and counterfactual structure are invariant to this substitution; only the point estimates and welfare m agnitudes will change.

Discussion

The reduced-form and structural results jointly establish five core findings. First, the transition from flat-fee public insurance to actuarially priced private coverage increased annual closure probabilities in the make-model comparison sample. Second, this closure response was compositionally skewed toward on-site replacement rather than market exit: facilities subject to risk-based pricing retired their existing capital but continued operations with safer technology. Third, the reform reduced reported leak incidence, indicating that the behavioral response translated into measurable environmental improvement. Fourth, the structural decomposition reveals that facilities are hypersensitive to explicit premium costs ($|\hat{\gamma}_{\text{price}}| > 1$) while fully internalizing private expected losses ($\hat{\gamma}_{\text{risk}} \approx 1$), implying that the remaining welfare distortion is attributable to the unpriced health externality rather than to behavioral undervaluation of insured risk. Fifth, the counterfactual comparison shows that a targeted mandate outperforms both the closure subsidy and the social optimum on net welfare, because it concentrates the intervention on the subpopulation where the externality wedge is binding.

The economic interpretation centers on the pricing mechanism documented across Sections 2 and 4. Under the flat-fee public fund, the marginal cost of retaining an aging tank is zero with respect to observable risk. Under the private market, the Mid-Continent rate schedule imposes discrete premium step-ups at actuarial age thresholds, generating a time-varying cost of continued operation that is increasing in precisely the risk characteristics that predict leaks. The make-model estimates capture the behavioral response to this newly introduced price signal among facilities for which the insurance reform was the sole contemporaneous policy shock.

The concentration of the closure response among high-single-wall-share facilities strengthens the pricing interpretation and is the premium-gradient-shaped pattern predicted by Proposition 3 and corroborated by the cross-validation evidence in Section 4.2 that observable risk characteristics genuinely predict outcomes. A general Texas-specific shock would produce a uniform shift in closure rates across all facility types. The observed heterogeneity, rising treatment effects for older and single-walled facilities, is inconsistent with a confounding shock and consistent with the specific cost differentials embedded in the actuarial rating algorithm.

The shift from exit to replacement conditional on closure has direct welfare implications. A policy that merely accelerates the shutdown of high-risk facilities achieves environmental gains at the cost of reduced economic activity in affected communities. The replacement margin documented here suggests that risk-based insurance achieves environmental improvement through capital upgrading rather than capacity destruction, preserving the service flow (fuel distribution) while reducing the associated externality. The structural welfare decomposition in Section 8 confirms this interpretation: the mandate, which targets the highest-risk subpopulation, achieves a positive net welfare gain precisely because it avoids the indiscriminate firm surplus destruction associated with the social optimum.

The structural finding that $\hat{\gamma}_{\text{risk}} \approx 1$ has a specific policy implication that the reduced-form estimates alone cannot deliver. If firms were myopic about environmental risk ($\gamma_{\text{risk}} \ll 1$), liability-based instruments such as higher statutory damage caps or expanded third-party standing would be the appropriate corrective. The data reject this hypothesis: firms correctly value private expected losses. The binding constraint is the health externality that falls outside the insurance contract. This means the most effective instruments are those that operate through the premium channel, including risk-adjusted surcharges calibrated to the externality, experience-rated deductibles, and Pigouvian taxes on observable risk characteristics, rather than instruments that rely on the threat of future liability.

Two caveats qualify the external validity of these estimates. First, the make-model sample by construction covers the range of risk profiles with common support across Texas and the control group. Facilities at the extremes of the risk distribution, very old single-wall tanks with no analogue in the control states, are excluded, and their behavioral response may differ in magnitude. Second, the Texas policy transition was a complete, overnight regime shift. States considering incremental reforms, such as risk-adjusted surcharges

within an existing fund structure, face a policy environment in which behavioral responses may be attenuated by partial premium adjustments or regulatory uncertainty.

Thirty-eight states currently operate flat-fee UST trust funds. The Texas evidence shows that observable facility characteristics, principally tank age and wall construction, generate predictable variation in environmental risk that private insurers can price, that firms respond to cost-based pricing in ways the theory predicts, and that the mechanism operates through capital upgrading rather than punitive exit. The path from the theoretical prediction to the empirical result runs through the pricing margin documented in the SERFF filings, validated in the cross-sectional risk models, and confirmed in the heterogeneous treatment effect by wall type.

Conclusion

This paper provides the first facility-level causal evidence on the impact of transitioning from a flat-fee public UST insurance fund to a risk-rated private market. The argument is built in four steps. First, the institutional record establishes that Texas's 1999 transition was abrupt, driven by fiscal insolvency, and immediately absorbed by a private market whose premium schedule conditions directly on age and wall type. Second, the theory shows that this premium structure, and only this structure, injects the facility's pollution hazard into its continuation value, generating a threshold-type closure rule that is absent under flat-fee pooling. Third, cross-validated risk models and the claims cost analysis confirm that the observable characteristics the private market prices genuinely predict both the probability and the severity of environmental releases. Fourth, the make-model comparison sample, constructed by matching on age bin and wall type, achieves balance on the dimensions along which the treatment effect is theoretically heterogeneous, earning parallel pre-trends in a way the unmatched comparison does not.

Using this design, the analysis demonstrates that forcing firms to bear the expected cost of their environmental risk profile has substantial effects on behavior. The introduction of risk-based pricing led to an immediate and sustained increase in the closure of high-risk, single-walled tanks. This response was characterized not by market exit but by technology upgrading, as firms chose to replace their obsolete tanks with safer alternatives. The behavioral change resulted in a substantial reduction in reported environmental leaks, concentrated among the facilities that the theory predicts should be most affected.

The structural estimation adds a dimension that the reduced-form estimates cannot provide: the decomposition of the behavioral response into a price channel and a risk internalization channel. The finding that facilities are hypersensitive to premiums ($|\hat{\gamma}_{\text{price}}| > 1$) while fully internalizing private expected losses ($\hat{\gamma}_{\text{risk}} \approx 1$) implies that the welfare distortion from flat-fee pooling is not a problem of firms failing to understand environmental risk. It is a problem of the insurance contract failing to transmit that risk into the firm's cost function. The counterfactual comparison confirms that instruments operating through the premium channel, particularly targeted mandates that concentrate intervention where the externality wedge is largest, dominate instruments that subsidize exit indiscriminately or rely on the threat of future liability.

The success of this price-based mechanism in reducing environmental risk invites a crucial policy comparison. Comparing the effects of risk-based pricing with vintage-differentiated regulations or direct technology mandates could illuminate the relative cost-effectiveness of these different policy instruments Shavell (2007). Such analysis would provide regulators with a more complete understanding of the optimal policy mix for managing the significant environmental hazards posed by aging infrastructure. Thirty-eight states currently operate flat-fee trust funds. The Texas evidence provides both the reduced-form proof of concept and the structural parameters required to evaluate the welfare consequences of reform.

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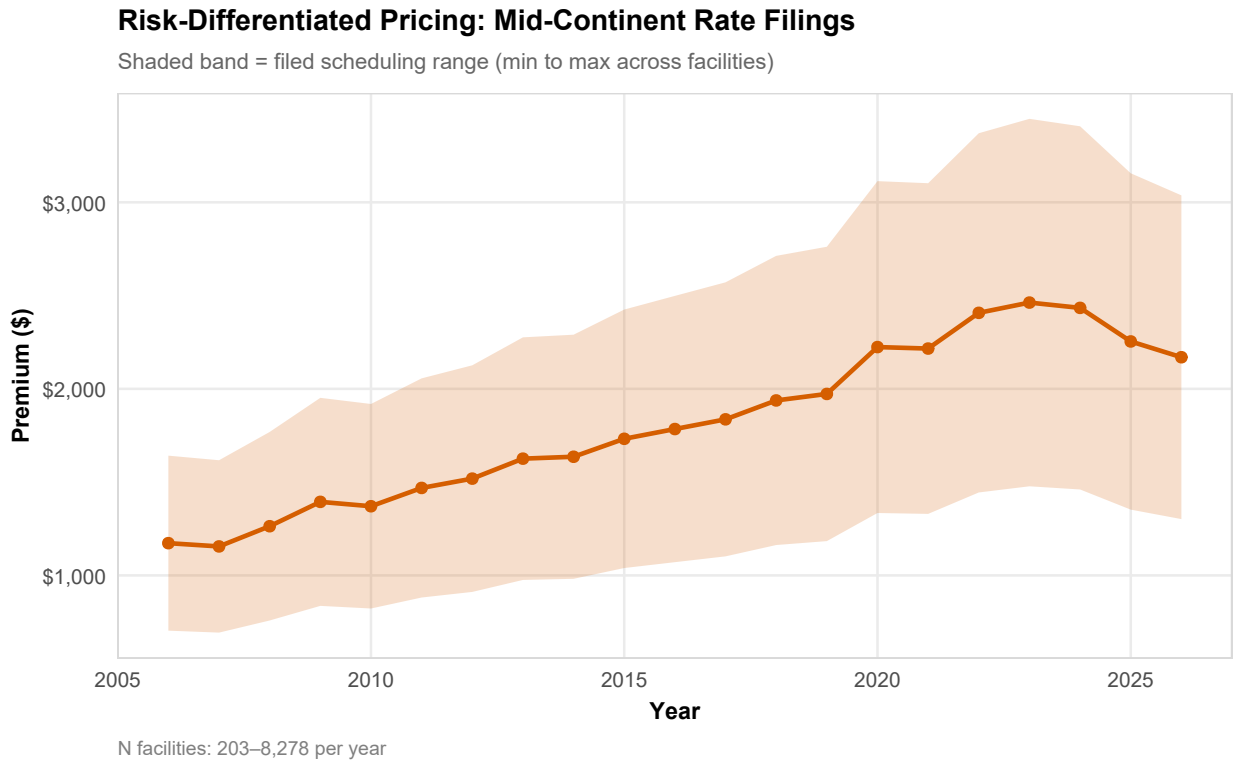
Appendix B: Data Appendix

Figure 15: Reconstructed Annual Premium Levels for Texas UST Facilities, 2006–2021.



Notes: Left panel: median premium by year, with pre- and post-transition shading. Center panel: premium index (1995 = 100) for single-wall and double-wall tanks separately. Right panel: log premium against tank age, with fitted lines by wall type. Premiums are strictly increasing in age and strictly higher for single-wall construction at every age, consistent with the Mid-Continent rate filing structure. Source: SERFF rate filings.

Notes: For each state, the table reports the total facility count, the number and share excluded due to missing tank installation or closure date records, and the treatment group indicator. Missingness rates are reported separately for Texas and each control state to facilitate visual inspection of differential attrition across treatment and control groups.

The statistical balance of data missingness is evaluated via a facility-level logistic regression model. Let M_i denote a binary indicator equal to 1 if facility i is excluded due to a missing tank installation or closure date, and 0 otherwise. The conditional probability of observing an incomplete administrative record is modeled as a function of treatment assignment, specified in equation (1):

$$\Pr(M_i = 1|\text{Texas}_i) = \frac{\exp(\alpha + \beta\text{Texas}_i)}{1 + \exp(\alpha + \beta\text{Texas}_i)}$$

where Texas_i is a dummy variable equal to 1 if facility i is located in the treatment state, and 0 if located in a control state. The coefficient of interest, β, captures the log-odds of data attrition for treated facilities

Table 15: Texas Phased UST Upgrade Mandate Schedule under 30 TAC Chapter 334.

[!h]

Vintage	Cohort	TX Deadline	Yrs Before Federal
On or before 12/22/1964	Pre-1965	12/22/1989	9.0
12/23/1964 – 12/22/1974	1965–1974	12/22/1990	8.0
12/23/1974 – 12/22/1979	1975–1979	12/22/1991	7.0
12/23/1979 – 12/22/1984	1980–1984	12/22/1992	6.0
12/23/1984 – 12/22/1988	1985–1988	12/22/1993	5.0

Notes: Each row corresponds to an installation vintage cohort. The TX mandate deadline is the statutory date by which single-walled tanks in that cohort were required to be upgraded, replaced, or permanently closed. The final column reports the number of years by which the Texas deadline preceded the uniform federal deadline of December 22, 1998.

Table 16: Unified Missing Data Summary by State and Treatment Group.

[H]

Table 17: Scale and Balance of Missing Tank Dates

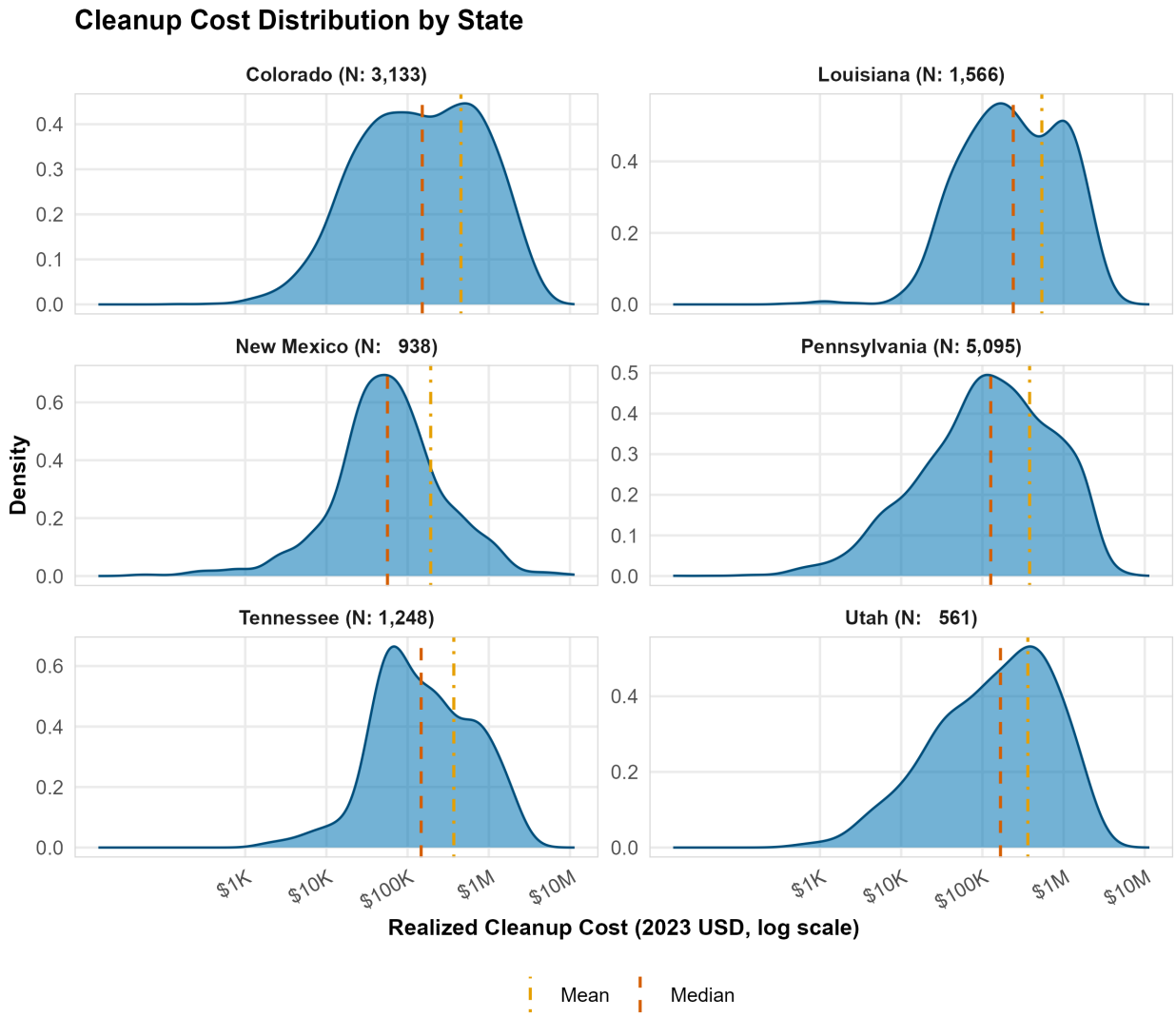
		Missing Install Date		Missing Closure Date		Overall Attrition	
Group	Facilities	Count	Rate	Count	Rate	Excluded	Rate
Control	232,848	53,648	23.04%	0	0.00%	53,648	23.04%
Texas	54,034	0	0.00%	0	0.00%	0	0.00%
Total	286,882	53,648	18.70%	0	0.00%	53,648	18.70%

Sample includes all incumbent facilities active prior to the statutory treatment year (1998). A facility is excluded if any associated tank lacks an installation date, or if a closed tank lacks a valid closure date. Logistic regression of missing installation date on treatment status yields p-value = 0.5361.

relative to controls. The maximum likelihood estimation yields a statistically insignificant coefficient on the treatment indicator ($p = 0.5361$). This result confirms the null hypothesis that $ $= 0$ $, establishing that the probability of exclusion is orthogonal to the regulatory mandate and satisfying the requisite identifying assumption for the primary difference-in-differences estimator.

Figure 16 presents the state-level distributions separately. The consistency of the heavy upper tail across all six states supports treating the skewed cost distribution as a structural feature of UST liability.

Figure 16: State-Level Distribution of Realized Cleanup Costs per Confirmed Release.



Notes: Deflated to 2023 dollars. Each panel plots the log-scale density for one state with vertical lines at the state-specific median and mean. The consistency of the heavy upper tail across all six states supports treating the skewed cost distribution as a structural feature of UST liability.